LEASING 15

LEASING IN THE AIRLINE SECTOR

Leasing plays a critical role in the financing of planes. IATA indicated that over 30% of the world's aircraft are owned by leasing companies. It was reported that lessors took delivery of 60% of all Airbus and Boeing planes delivered in 2021. The availability of lease financing is a double-edged sword, as one effective barrier to entry – the capital cost of planes – is reduced, which then leads to increased competition from new entrants. Sometimes this may lead to success. For example, Branson started Virgin Atlantic with one plane to fly the trans-Atlantic route. However, many airlines do fail and the leasing companies take back the leased planes and lease them out to the next set of hopefuls.

The commercial life of planes is about 25–30 years if we include the freight market. The purchase of planes using loan finance is expensive as airlines are required to make a pre-delivery payment (PDP) of about $15–20 million per new plane from Boeing or Airbus, 2–3 years prior to delivery. Owning a plane means the company is required to finance the PDP and the capital cost with loan finance or internal cash resources. Loan financing may result in liquidity problems for the company whilst leasing offers greater flexibility. Also, ownership requires the airline to take on the residual risk of the planes. The introduction of IFRS 16 in 2019, which requires the capitalisation of operating leases, has significantly affected the airline industry and has resulted in higher debt-to- equity ratios, higher EBITDA margins and lower returns on capital for airlines that predominantly lease aircraft. Prior to IFRS 16, airlines that leased planes in terms of operating leases, did not include planes on their statements of financial position and leasing became known as a form of off- balance sheet financing. Airlines were the one of the most affected industries by the capitalisation of operating leases. Leases have to be paid and COVID had a catastrophic impact on demand and yet lease payments were often required to be made to lessors. This combination of the collapse in demand and high fixed costs led to the demise of Comair (BA in South Africa) and Kulula.com.

In this chapter we will describe the types of leases, the accounting for leases, and the impact that capitalising leases will have on the Statement of Financial Position. We will evaluate the reasons why companies lease rather than purchase assets and we will evaluate leasing decisions relative to borrow and purchase decisions. Finally, we will explain and apply the adjusted present value approach to evaluate capital projects with financing tax shields.

LEARNING OBJECTIVES

After working through this chapter, you should be able to:

- Outline the various types of leases and how each type of lease provides the right to use an asset.
- Identify the impact that IFRS 16 will have on the accounting for leases and on financial ratios.
- Describe a direct lease, sale and lease-back and a leveraged lease.
- Identify the advantages of leasing.
- Evaluate the leasing decision and alternative sources of finance using discounted cash flows by being able to:
  – select and justify an appropriate discount rate; and
  – calculate the net present cost of the various financing alternatives and the net advantage of leasing.
- Calculate the impact of an assessed loss on the financing cash flows.
- Determine the adjusted present value of a project.

INTRODUCTION

A lease is a contract that conveys the right to the use of assets for a set period of time in exchange for a periodic payment. The lessee enters into a lease agreement promising to make lease payments in return for the right to use an asset. The lessor, who is the legal owner of the asset, promises to supply the asset in exchange for lease or rental payments. As a lease entitles the lessee to the use of the asset, it is appropriate to consider leasing as a source of finance. It is worthy of special attention

as a number of the financing issues are particular to leasing. The lease rental or payment may be paid at the beginning of each month, in the period covered by the agreement, and is included in the gross taxable income of the lessor. On the other hand, the lessee would be allowed to deduct the lease payment in calculating its taxable income. These provisions may enable tax deductions to be transferred from the lessee to the lessor. At the termination of the lease the lessee may have the option of renewing the lease, acquiring the asset, or returning it to the lessor. These options would be specified in the original agreement. Leasing is a widespread form of medium-term and long-term financing and the growth in lease financing has been extraordinary. This chapter explores the use of a discounted cash flow approach using the cost of debt as the relevant discount rate to evaluate leasing decisions.

In the chapter on capital budgeting (Chapter 8), it was assumed that the cash flows discounted in the evaluation of an investment proposal depended on the firm's ownership of the asset in question. Thus, the cash flow analysis contained an initial outflow at the beginning of the first period, which was the initial purchase price. However, the firm could decide to lease the asset rather than buying it outright. To take account of such a situation, the capital budgeting model could be changed to incorporate cash flows associated with leasing rather than outright purchase. This would result in financial managers viewing the leasing of an asset as an alternative to buying the asset. Only the financing and taxation cash flows will change, the other cash flows do not change as they stem from the use of the asset in operations.

Leasing an asset will be evaluated as an alternative to borrowing and purchasing the asset. In relation to the banks, leasing is equivalent to borrowing and if a company decides to lease an asset, it effectively is using up part of its debt capacity. This chapter will explain the accounting of leases in terms of IFRS 16 and will evaluate leasing as an alternative form of finance.

1 TYPES OF LEASES

The three most important types of leases are *direct leases, sale and lease-back transactions* and *leveraged leases*. The distinction among leases on this basis can be significant to both parties, as they will alter the cash flow pattern of both the lessor and the lessee.

Direct lease

In a direct lease, the lessor acquires an asset from the manufacturer in order to lease it to the lessee. A direct lease is the most straightforward type of lease, and also the type most frequently encountered in practice. A lease that results in the lessee having the use of an asset for its useful life would be considered a direct lease.

In some instances, the manufacturer may itself act as the lessor. This type of lease can be used to aid marketing campaigns. Low interest rates allow the manufacturer to promote the product, subsidising the cost of finance incurred by the customer, without reducing the selling prices. In this chapter we will focus on direct leases.

Sale and lease-back

A sale and lease-back form of lease can be employed in connection with existing or new assets. A firm sells an asset under an agreement to lease it back from the buyer. In exchange for the asset the selling firm obtains cash, and as the future lessee, the long-term right to the use of the asset. On the other hand the lessor pays out cash but obtains the legal right to the asset and the right to lease instalments. At the end of the lease agreement there may be provision for the lessee to repurchase the asset at a specified price.

Sale and lease-back is commonly used by firms that own assets and need to raise funds. Frequently the assets are specialised and, in this manner, a firm can obtain funds through the sale of the assets without losing the right to use such assets. Further, if specified in the agreement, the firm can re-acquire the asset at the end of the period of the lease.

Leveraged lease

With a leveraged lease, the lessor does not provide all the funds required to purchase the asset that is to be leased. The balance of the funds would be obtained from other financial institutions and the lease payments made by the lessee are then used to service these loans. Any excess of the lease payment after the loan coverage is retained by the lessor. An attraction of the leveraged lease, for the lessor, is a potential for profits provided by the use of financial leverage and the tax deductions. However, the participation of companies from other sectors in leveraged lease structures to take advantage of the tax deductions has been curtailed as a result of the introduction of Section 23A of the Income Tax Act. This limits deductions such as the depreciation allowances to the rental income of the firm. Normally the lessor's risk is not limited to the lessor's own investment. Should the lessee default, the lessor has a financial

obligation to the other financial institutions that have lent funds. In general, leveraged leasing is used to finance large projects. Figure 15.1 depicts the structure, cash flows and entities involved in a leveraged lease.

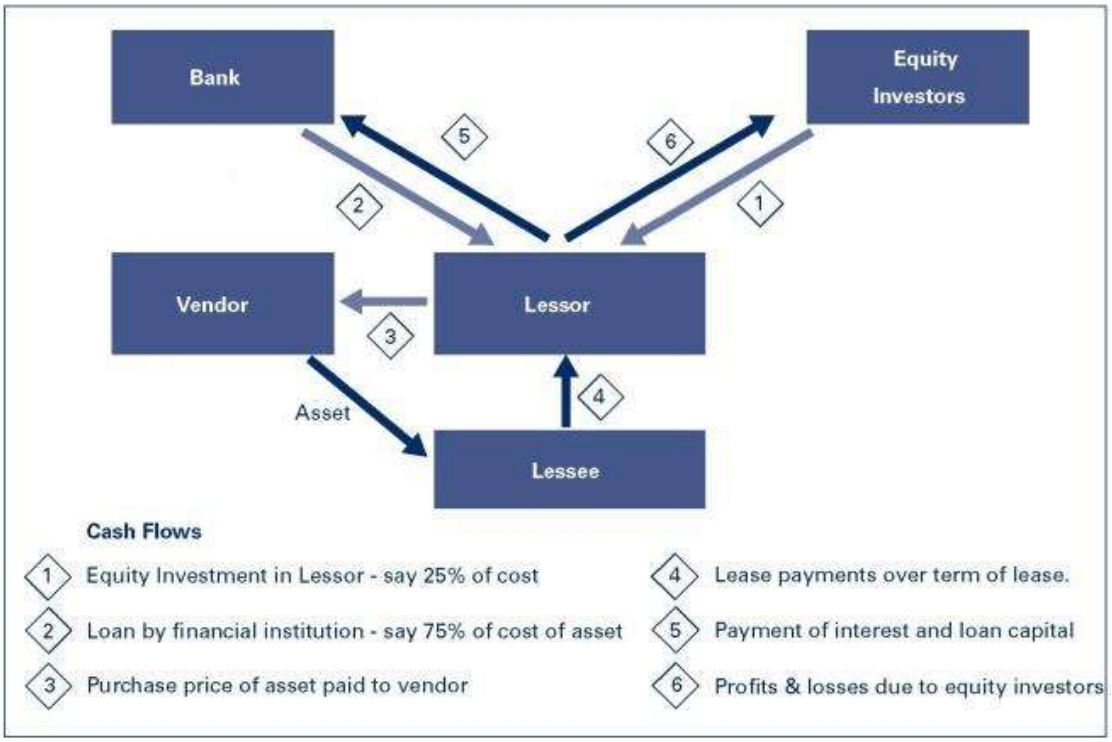

Figure 15.1 The cash flows and entities involved in a Leveraged Lease

While these different types of lease differ in complexity and administrative detail, we can evaluate each type of lease as an alternative source of finance.

2 OFF-BALANCE SHEET FINANCING AND LEASING

Leasing often provided "off-balance sheet" financing. In fact this was promoted as a major advantage of leasing. However, if a lease falls within the definition of a lease in terms of IFRS 16, then the company is required to bring the leased asset onto the Statement of Financial Position. It does this by disclosing a right-of-use asset and the present value of the Page 15-4 future lease payments as a liability. In terms of IFRS 16, if a contract is defined as a service contract or if it does not meet the definition of a lease, then the payment is expensed as incurred and there will be no asset or lease liability.

What do we mean by "off-balance sheet" financing? As the firm does not own the asset, the asset and the liability reflected in the contract may not be reflected in the Statement of Financial Position. Let's see the effect that a financing structure may have on the Statement of Financial Position of a company considering a decision to

incur a cost of R400m for plant and equipment. The Statement of Financial Position prior to the decision to invest and after is shown below:

		Before	After
Statement of Financial Position	**Rm**	**Borrow & Buy**	**Lease**
Non-Current assets	400	800	400
Current assets	300	300	300
	700	1,100	700
Equity	350	350	350
Long-term loans	350	750	350
	700	1,100	700
Debt/Equity	100%	214%	100%

The capital structure of the firm prior to the expansion reflects a debt/equity ratio of 100%. If the company decides to invest R400m in fixed assets and finances this with a loan, then the Statement of Financial Position will reflect the increase in assets and the increase in the debt liability. This results in a significant increase in financial leverage with the debt/equity ratio increasing to 214%. If the contract, which includes access to plant and equipment, is not defined as a lease in terms of IFRS 16, it may not be required to reflect this expansion on the Statement of Financial Position even though the company has entered into a contract requiring the firm to make contractual payments for a number of years. The risk profile of the company has clearly changed but this is not reflected in the financial leverage ratios of the company.

What about the Statement of Comprehensive Income? If the contract is not a lease, then the full contract payment will be reflected as an expense for the year. If it is defined as a lease, then the company will reflect a depreciation charge on the asset and the lease payment will be divided into a capital and interest component. The implicit interest component is deducted as an expense from income. IFRS 16 has made it much more difficult for firms to avoid bringing the lease commitments and a right-of-use asset onto the balance sheet. Time will tell whether firms will be able to structure contracts in such a way as to ensure that a contract is not defined as a lease, which will enable off-balance sheet financing. We will now analyse in greater detail the accounting for leases in terms of IFRS 16.

3 THE ACCOUNTING FOR LEASES IN TERMS OF IFRS 16

The accounting for leases changed in 2019 with the implementation of IFRS 16. This has been called one of the most momentous changes in accounting standards history. It was estimated by regulators that IFRS 16 would bring $3 trillion of leasing

commitments onto balance sheets. This is a big number and materially increased reported debt as well as increasing the associated right-of-use assets. However, the net effect was to materially and profoundly increase the debt/equity ratios of companies in sectors such as retail, logistics, hotels, airlines, consumer discretionary and healthcare.

It is our view not only that the accounting for leases changed but also that we may see a rearrangement of the commercial terms of contracts to mitigate to some extent the ambit of IFRS 16. IFRS 16 defines a lease as "a contract or part of a contract that conveys the right to use an asset (the underlying asset) for a period of time in exchange for consideration". IFRS 16 requires the recognition of a lease liability which represents the present value of future Page 15-5 non-cancellable lease commitments as well as a right-of-use asset. IFRS 16 leads to significant impacts on financial metrics and ratios and we have set out some of these effects in Chapter 5. So, how did we get here?

The prior lease accounting standard, IAS 17, focused on the definition of finance leases and operating leases. If a lease was economically similar to purchasing the underlying asset, then the lease would be classified as a finance lease and would be reflected on a firm's balance sheet. Specific rules and criteria were applied to determine whether a lease was a finance lease. All other leases were classified as operating leases, which were accounted for in the same way as other service contracts and resulted in a periodic rental expense. The right to use an asset in terms of an operating lease had no effect on the balance sheet. In contrast, finance leases resulted in right-of-use assets and lease liabilities on the balance sheet. Operating leases became known as "off-balance sheet financing".

The difference between finance and operating leases should reflect that in the case of finance leases, the risks and rewards of ownership were effectively transferred to lessees. Yet IAS 17 failed to adequately reflect the economic reality that leases represent long-term financial liabilities. It was difficult to compare companies which owned their assets and companies that leased their assets. Rating agencies and analysts would make their own adjustments and estimate the present value of lease obligations by multiplying the annual lease payment by 8, although perhaps this was too conservative and a 5–6× may be more appropriate. Depreciation was estimated to be 2/3 of the annual charge and the balance was assumed to be interest. These were simple rules of thumb and short-cuts trying to solve a complex problem about a firm's financial obligations.

Avoiding the definition of a financial lease was so easy that a major logistics provider was able to report an operating lease charge of R1.5bn and a finance lease charge of R7m in a particular year. Logistics providers and airlines all made extensive use of "operating leases" in order to enhance returns and reduce debt ratios. The former chairman of the IASB, Sir David Tweedie, was quoted as saying, "One of my great ambitions before I die is to fly in an aircraft that is on an airline's balance sheet". What can one say? A true accountant to the end. IFRS 16 will mean that his "great ambition" has come true.

IFRS 16 means that there is a single lease accounting model and that there will no longer be a lease classification test of whether a lease is an operating or a finance lease. All leases are on the balance sheet, which means that a lease is accounted for in a similar manner as a purchase of an asset which is financed with a loan. IFRS 16 applies to all leases except leases that are 12 months or less or for assets that are of low value (USD 5 000 or less). Given these exemptions, the next key question is whether a contract represents a lease. We will come back to this issue later.

The effect of IFRS 16 on financial statements, ratios and processes

How does IFRS 16 affect a company's accounting for leases and the numbers it reports in its financial statements? The following are some of the effects of IFRS 16.

- Recognition of a lease liability based on the present value of future non-cancellable lease obligations based on the rate implicit in the lease or the firm's incremental borrowing rate
- Recognition of a right-of-use asset initially equal to the lease liability, adjusted for prepayments and direct costs, which reflects the value of the right to use the underlying asset over the lease term
- The lease payment is divided into two components – an amortisation of the lease liability over the term of the lease and an interest component which is reflected in the income statement as an interest expense
- The right-of-use asset is amortised/depreciated in accordance with the firm's depreciation policy. A straight-line depreciation over the term of the lease will mean a higher periodic combined interest and depreciation expense in the early years as compared to the later years of a lease.

 What are the effects of IFRS 16 on financial ratios? The implementation of IFRS 16 had the following effects on financial ratios and numbers:

- Increase in EBITDA and a firm's EBITDA margin
- Increase in the debt/equity and debt ratios
- Lower asset turnover ratios
- Increase in EBIT and EBIT margin
- Decrease in rental expense and increases in depreciation and interest expenses
- A lower net income in the early years of a lease and a higher net income in later years
- Lower ROCE and ROIC ratios due to a higher asset base
- Lower current ratio.

IFRS 16 may affect loan covenants, particularly when it comes to the renewal of loan facilities. Initially loan covenants should include a provision that loan covenants are based on the accounting standards at the time loans were undertaken.

Lessees are required to identify all lease contracts, and this will require a careful analysis of all contracts in place that involve the use of assets. Lease management IT systems are necessary to maintain the required data on each lease and track annual assessments and values over time. It has been estimated that a company will need to capture up to 100 data fields per lease such as payments, expenses, termination options, interest rate, renewal options, purchase options and residual value guarantees, if applicable.

IFRS 16 has increased the volatility of balance sheet values, as the standard requires that key estimates be assessed in each reporting period. If leases are US Dollar-based, as airplane leases mostly are, then there will be an increase in the reported exposure to foreign currency risks.

Is a contract a lease or does the contract contain a lease?

IAS 17 required that we determine whether a lease is an operating or a finance lease. IFRS 16 requires that we shift to questions such as whether the contract is a lease or a service. Does a contract constitute or contain a lease? If not, we then carry on as before and the payment is a periodic expense. The definition of what a lease is becomes key to the accounting for leases.

The flowchart set out in Figure 15.2 is derived from a more detailed flowchart in IFRS 16, and sets out the steps involved in the assessment of whether a contract is or contains a lease.

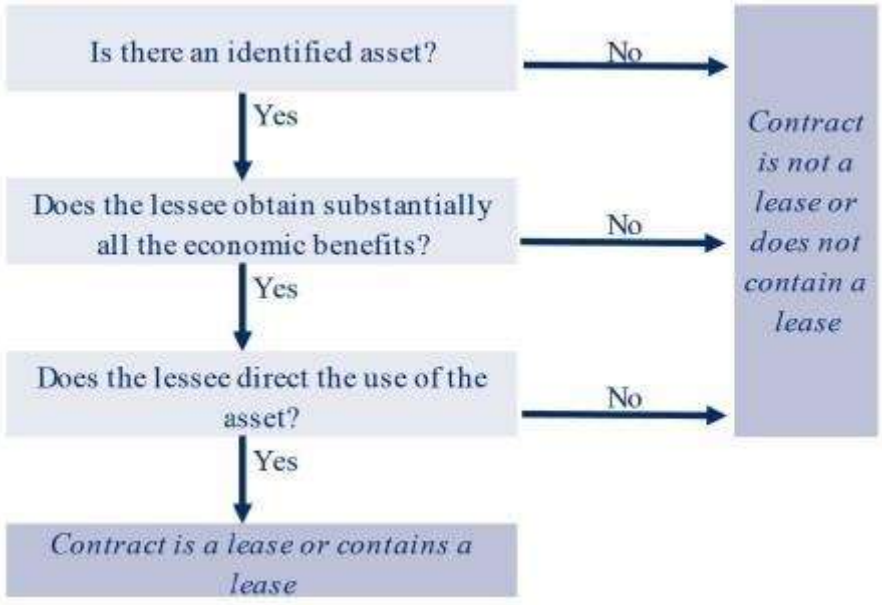

Figure 15.2 Flowchart to assess whether a contract is or contains a lease Page 15-7

Is there an identified asset?

For a contract to be or contain a lease, there needs to be an identified asset which is explicitly or otherwise specified in the contract. The lessee needs to control the asset, for if the lessor has a substantive right to substitute the asset for another asset, then a lease does not exist in terms of IFRS 16. In order to be substantive, there should be economic benefits of substitution for the lessor. A lease can be a portion of an asset if it is distinct, such as a separate floor or section of a building. A capacity portion may be an identified asset if it represents substantially all the capacity of the asset.

Does the lessee obtain substantially all the economic benefits from the use of the asset?

This will apply if the benefits of operating the leased asset flow to the lessee in terms of what the asset produces and the economic value generated by the asset. This will be based on fact, and the economic benefits should be derived from the *use* of the asset. IFRS 16 indicates that the inclusion of a variable lease payment based on turnover does not mean that the lessee does not obtain substantially all the economic benefits from the use of the asset. However, this requirement will mean that contract terms and the economic facts need to be analysed. If a lessor of a CT scanner receives income

only on use, then it is an open question whether the lessee obtains substantially all the economic benefits from the use of the asset.

Does the lessee direct the use of the asset?

This requirement will be met if the lessee has the right to direct how and for what purpose the asset will be used over the term of the lease. The lessee should have decision-making rights about the use of the asset to enhance the economic benefits derived from the use of the asset. This may involve decisions with regard to outputs from the asset, timing (when the asset is used) and location (where it is used) and the right to cease production. If there are predetermined conditions on use, then the lessee will have the right to use the asset subject to these conditions or operating instructions, but without the lessor having the right to change the conditions or operating instructions during the term of the lease. This will require an analysis of the decision-making rights set out explicitly or implicitly in a contract. Which party controls the day-to-day operations of the asset may be relevant. IFRS 16 has a threefold classification of decision-making rights:

The ability to make "how and for what purpose" decisions will indicate whether there is a lease unless this is predetermined, in which case we follow whether the lessee is able to make day-to-day operating decisions. Protective rights may curtail the use of an asset by the lessee but will not on their own prevent the lessee from having the right to direct the use of an asset. Protective rights may state that a lessee is not permitted to transport hazardous materials, or that the asset is required to be serviced at specified intervals, or that planes cannot fly over conflict areas. These rights protect the lessor's investment in the asset and will not generally affect the ability by the lessee to direct the use of the asset.

> *Example: A lessee has a contract regarding the use of a fleet of trucks over a period of time. Even if the lessor operates and maintains the trucks, the contract is a lease if the lessee has the right to decide how, where and when the trucks are used.*

Non-lease components

Contracts may include both lease and non-lease (service) components. For example, a contract for a truck which includes a lease may also include maintenance services. The lessee will need to allocate a portion of the consideration to the non-lease

component based on the standalone price of the non-lease component. Only the lease component will be included in the right-of-use asset and associated lease liability.

Lease term, lease payments and the discount rate

The key issues of a lease relate to the lease term, the lease payments and the discount rate used to value the future lease contract payments. We explain these terms as follows:

Lease term	Lease payments	Discount rate
• The lease term is the non-cancellable period of the lease plus any period included in any option to renew if it is reasonably certain that the lessee will take up the option to renew and extend the lease period plus any period covered by an option to terminate the lease if it is reasonably certain that the lessee will not exercise that option	• Lease payments refer to payments made by the lessee relating to the right to use an underlying asset during the lease term and include fixed payments, variable lease payments that depend on an index or a rate (such as CPI), the exercise price of any purchase option if the lessee is reasonably likely to exercise the option plus residual value guarantees	• The lease payments should be discounted using the interest rate implicit in the lease or the lessee's incremental borrowing rate if the interest rate implicit in the lease cannot be readily determined.

Lease payments do not include payments allocated to non-lease components. Lease payments do not include variable lease payments linked to the performance of the underlying asset. Lease incentive payments by the lessor should be set off against the lease payments.

The **implicit rate** is defined as "the rate of interest that causes the present value of the lease payments and the unguaranteed residual value to equal the fair value of the underlying asset and any direct costs of the lessor". The **incremental borrowing rate** is defined as "the rate of interest that a lessee would have to pay to borrow over a similar term, and with a similar security, the funds necessary to obtain an asset of a similar value to the right-of-use asset in a similar economic environment".

Structuring considerations

In terms of IAS 17, firms structured the terms of a lease in order to avoid the definition of a finance lease. In terms of IFRS 16, there may be ways to structure a contract so that it is not defined as a lease or so that the lease component of any contract is minimised. The objective is to reduce the lease liability and the right-of-use asset on the balance sheet.

Lessees may request that the terms of leases be reduced in order to reduce the value of the lease liability and right-of-use asset (and we expect this to happen), but this may result in higher risks for lessors, particularly if the term of the lease impacts on the financing options of lessors and for lessees in relation to renewal risks. Variable lease payments will be expensed as incurred and will not be subject to IFRS 16 if these payments are not in-substance fixed or depend on an index or rate.

Judgement may be required to determine the reasonable certainty that an option to renew will be exercised. Uncertainty about exercise will reduce the lease term and the value of a lease liability and associated right-of-use asset.

It is expected that lessee firms will renegotiate lease terms in order to reduce the effect of IFRS 16. This may include reducing the term of a lease, which will reduce the present value of the lease liability. It may result in the separation of components such as maintenance from any lease payment, which up to now may not have been separately identified. This will reduce the lease payment and therefore the value of the lease liability. Firms may wish to include variable components (not based on a rate) which will not be included in the valuation of leases. A variable component will also reduce the risk to a firm but may increase the risk to a lessor, although the lessor may price this into the lease contract.

However, there are real risks of changing the terms of leases in order to minimise the ambit of IFRS 16. The structuring of contracts to avoid the lease definition may impact operations and business flexibility and may lead to legal risks if economic incentives change during the term of the lease or the period of the expected use of the asset.

Application of IFRS 16: an example

B Ltd has financed the acquisition of plant and equipment with a lease which requires an annual lease payment of R150m, payable in arrears over the next 4 years. The current date is 30 June 20x1. The operating lease charge of R150m would have been included as an expense in the 20x2 financial year under the previous standard, IAS 17. You wish to understand the impact that IFRS 16 has on the earnings of the company and statement of financial position. The interest rate charged by the lessor reflects the market rate and is 12% per year.

Step 1: Value the right-of-use asset

If we discount the lease payments at the incremental borrowing rate of 12%, the present value of the future lease payments is R455.602m and this represents the value of the right-of-use asset.

	0	1	2	3	4
PV of lease payments as at 30 June 20×1	**20×1**	**20×2**	**20×3**	**20×4**	**20×5**
Lease payments		150	150	150	150
PV of lease payments as at 30 June 20x1	455.602				

Step 2: Determine the interest expense and the liability at amortised cost using the effective interest method

We set up an amortisation table as follows:

Year	Open balance	Interest	Pmt	Interest + Pmt	Closing Balance
20x2	455.602	54.672	-150.000	-95.328	360.275
20x3	360.275	43.233	-150.000	-106.767	253.508
20x4	253.508	30.421	-150.000	-119.579	133.929
20x5	133.929	16.071	-150.000	-133.929	-0.000

For example, for the year 20x2, the company will now report an interest charge of R54.672m and the liability at year-end will be R360.275m.

Step 3: Determine the amortisation of the right-of-use asset

If we assume the straight-line amortisation, of the right-of-use asset, then the depreciation (amortisation) expense in each year will amount to: R455.602/4 = R113.9006m.

The value of the right-of-use asset at the end of 20x2 will be: R455.602m – R113.9006m = R341.7018m.

What is the impact of IFRS 16 on the Statement of Comprehensive Income? In our example, we deduct the amortisation (depreciation) expense and deduct the imputed interest expense as part of the firm's total interest expense. The pattern of total expense (amortisation + interest) relative to the annual lease payment over the four years is presented in Figure 15.3. Therefore, Page 15-10 IFRS 16 reduces net income in the early years of the lease and increases income towards the end of the lease term. In the short term, IFRS 16 has a negative impact on net income relative to the pre-IFRS 16 treatment of operating lease payments. The cash tax charge will not change as the operating lease payment will continue to be tax deductible as and when incurred. There will be deferred tax implications but this will have little relevance for analysts.

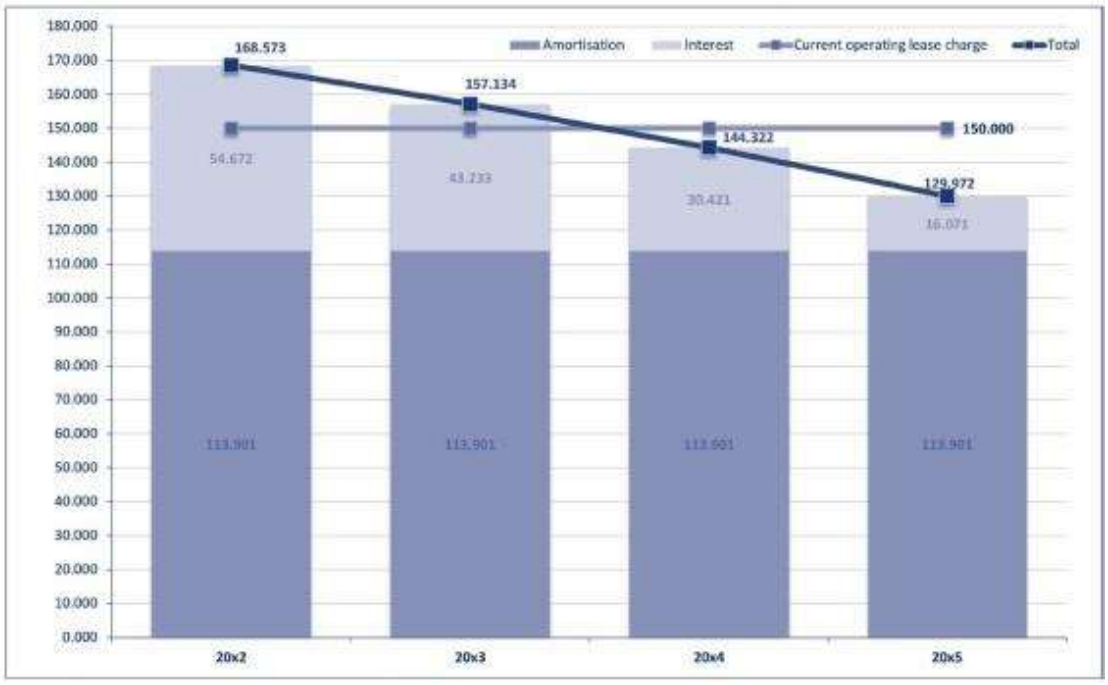

Figure 15.3 Total expenses: IFRS 16 versus operating lease expense in terms of IAS 17

Although the total expense when we add amortisation to the interest expense will be higher than the lease payment in 20x2 (and 20x3), the fact is that only the amortisation expense will be included in operating costs. We will therefore see an increase in operating income (EBIT) in 20x2 of R36.099m (R150m – R113.901m) as the interest expense of R54.672m in 20x2 will now be moved to financing expenses.

4 ADVANTAGES OF LEASING

In many instances a firm will have no choice regarding the decision to lease or to buy an asset. In certain circumstances a manufacturer may wish to retain control over the disposal of an asset. In such an instance the manufacturer would not be willing to sell the asset but would be prepared to lease it. In many other cases it makes good business sense to lease. For example, a retailer may find it attractive to open a store in a shopping complex. As retailers are not normally in the habit of financing the building of shopping complexes, they would need to rent the space for their store. There are a number of reasons why leasing is popular.

Changing technology

In some industries the rate of technological change is rapid, so an advantage of leasing is that the term of a lease can be considerably shorter than the life of the leased asset. Alternatively, a long-term lease may have a cancellation clause that can be exercised at the option of the lessee. This enables the lessee to avoid the risk of an asset

becoming obsolete or redundant prior to its originally estimated useful life. This is particularly applicable to assets employing advanced technology.

If the lessor is forced to bear this risk, then the lessor would seek compensation through higher rentals. However, much of the lessee's risk would not be transferred to the lessor as it would stem from redundancy. Redundancy would be caused by the changing needs of the lessee, and so, while an asset may have outlived its usefulness for that particular lessee, the lessor on the other hand will be able to find another firm that is prepared to lease the asset. Page 15-11

It is also possible that the expected cost associated with obsolescence will be lower for the lessor than for the lessee. This could occur through the lessor's ability to predict change and by having a ready market for obsolete assets. Thus if the lessor has greater opportunities for either obsolete or redundant equipment, the lessee would realise real benefits from leasing rather than buying. Equipment manufacturers and IT firms will lease equipment to customers and will often permit firms to upgrade to take advantage of new technology. The equipment manufacturer will often have a leasing subsidiary and will permit customers to cancel existing leases when upgrading to new equipment thereby replacing an existing lease with a new lease. The company is not locked into old technology and the manufacturer is able to increase sales of its new products and equipment.

Tax advantages

Under certain conditions a lease can result in lower combined taxation paid by the lessor and the lessee relative to the tax paid if the lessee were to buy an asset. For example, if a firm has an assessed tax loss, or does not expect sufficient income to utilise the depreciation deductions and interest tax shield generated by owning plant, then by leasing the plant from a profitable lessor, the lessee can transfer these tax shields to the lessor. The lessor would then lower the required lease payments and, in effect, the savings would be shared between the lessee and lessor. Section 23A of the Income Tax Act may limit the depreciation deductions to taxable rental income. This is to ensure that firms that normally do not act as lessors do not engage in structures to become lessors simply to take advantage of the depreciation deductions.

Obtaining 100% debt financing

In obtaining 100% debt financing, the major motivation of the lessee is to acquire the use of an asset without having to provide any cash investment as an alternative to

paying a deposit and purchasing the asset on an instalment sale.

A firm may find itself in a situation where it is expanding rapidly and, as a result, has a cash flow problem. In such an instance, leasing would be preferable to borrowing as, with borrowing, the firm would be required to invest some of its funds in acquiring the use of the assets. Similarly, a firm wanting to increase its financial leverage will be able to do so more effectively through leasing than through borrowing, for the same reason.

Operating flexibility

Companies may be able to react more quickly to changes in market conditions. For example, an airline may change routes more quickly and more cost effectively if it is able to lease the right type of aircraft for certain routes. If particular routes do not make financial sense but other routes open up which are more profitable, requiring a different type of aircraft, then the airline will be able to maximise operational flexibility by leasing rather than owning aircraft.

Reduction in operating leverage

In some cases and for particular assets, a company may be able to enter into an agreement that requires the company to pay an amount per use rather than per period. This will result in changing a fixed cost to a variable cost and lessors are able to charge a greater fee. The lessor may be able to diversify its risk as it is leasing its assets to many customers.

Coping with uncertain demand

A company that is uncertain of demand for its products is able to reduce its exposure to fixed costs and long-term commitments by undertaking a lease with a cancellation clause. However, if demand increases substantially, then leasing enables the company to increase supply by leasing equipment.

Specialisation effects on maintenance, residual values and purchase costs

A specialised leasing company will be able to negotiate lower purchase costs and thereby achieve savings due to its purchasing power. A leasing company may be purchasing 20 trucks at a time and therefore is able to obtain lower prices. Furthermore, the company will be able to obtain higher resale values for used

equipment. Also the company will be able to contract more efficient maintenance of its equipment and vehicles owing to economies of scale.

Standardisation of contracts

Leasing is subject to standardised contracts meaning that it is easier to arrange from an administrative perspective.

Fewer restrictions

Leasing may involve fewer restrictions than loans as the lessor's security is the asset and the lessor understands the value and risk of the leased asset. This means that the lessor will require less financial and credit information.

Off-balance sheet financing

In terms of IFRS 16, companies are required to bring all leases onto the balance sheet (Statement of Financial Position) by including a right-of-use asset and a lease liability, which is the present value of future lease payments. However, a company may have the flexibility to reduce assets by reducing the term of the lease, thereby reducing the right-of-use asset and lease liability in relation to ownership. For example, the right-of-use asset for a 5-year lease will be significantly less than the value of the asset which has a 20-year useful life. Furthermore, payments to the lessor may include non-lease components, such as maintenance or lease payments which are variable, or the lessor may retain control, or there may be no identified asset. These components or attributes would imply that the related payments will not constitute leases subject to IFRS 16.

Avoidance of capital expenditure controls and budgetary constraints

Companies may impose more stringent requirements on capital budgeting than on leasing decisions. Also, budgetary constraints will encourage the leasing of assets by companies and in particular by public sector agencies. A study in the UK on why lessees prefer to lease rather than borrow indicated that the leasing decision is not based purely on choosing the lowest cost alternative or financial considerations. The study indicated that firms find leasing attractive for the following reasons:

1. Leasing is easier to arrange from an administrative point of view than borrowing.
2. Lease finance can be obtained with greater ease and fewer restrictions than other forms of finance.

3. Leases can be arranged in which the rental payments increase over the lease period, thus enabling low rentals to be charged against profit in the early stages of a project's life.
4. Leasing provides a way to avoid capital expenditure controls.

It is debatable whether all of the reasons given would add value to the firm. For example, the circumvention of capital expenditure controls could have exactly the opposite effect by distorting the effective allocation of capital within the company. There are other disadvantages as well such as: loss of flexibility of use such as the inability to close an unprofitable store subject to a long-term lease, lack of ownership, no long-term security over location if this is important to value and escalation clauses in property leases. Page 15-13

5 EVALUATING THE LEASING DECISION

There has been considerable debate as to whether the decision to lease is an investment decision or a financing decision. The two schools of thought are based on the premise that the evaluation is 'lease versus purchase' or 'lease versus borrow' respectively. The view taken in this text is that the decision rests on a choice of leasing or borrowing and buying an asset, which is a financing decision. Thus the evaluation will be based on an assumption that the investment proposal has already been accepted. Since leasing is the functional equivalent of debt financing, it represents a direct alternative to borrowing funds for the outright purchase of an asset. The attractiveness of a lease proposal should thus be evaluated in terms of the present value of the differential cash flows relative to the borrow-and-buy alternative. It should be noticed that, once again, we are about to evaluate cash flows using the present-value technique so as to take into account the time value of money. To conduct such an analysis we need to identify firstly an appropriate discount rate, and secondly the relevant cash flows for the analysis.

Selecting an appropriate discount rate

It will be recalled from the chapters on capital budgeting that the discount rate that was considered appropriate with the investment decision was the weighted-average cost of capital. In certain instances it was necessary to use either a higher or lower discount rate depending on the risk associated with the individual project. For example, if the project being evaluated had a higher risk than the average for the firm,

then a higher discount rate would be used, while the converse would hold if the project had a lower risk.

With a leasing decision, management are examining a financing decision and so the appropriate discount rate is the after-tax cost of debt. As the cash flows associated with leasing would be much more certain than those associated with the investment decision, a lower discount rate is appropriate. Thus a general rule emerges that, when analysing investment cash flows, the weighted average cost of capital is used as a discount rate, and when analysing financing cash flows, the after-tax cost of debt is the appropriate discount rate.

Some cash flows are common to both the investment decision and the financing decision. For example, in the investment decision it is necessary to take account of the depreciation deductions in arriving at a present value. In evaluating the cost of leasing, it will be necessary to establish the cost of these foregone tax shields. Thus it may seem strange that in one model the depreciation deductions are discounted at the weighted-average cost of capital, and in the other model the allowances are discounted at the after-tax cost of debt. Obviously the riskiness of this cash flow does not change from one model to the other, so which discount rate is correct? Conceptually it is preferable to use the lower discount rate as tax allowances and depreciation deductions are regarded as being relatively certain. However, it is unnecessary to use multiple discount rates within the capital budgeting model as the weighted-average cost of capital which reflects the return required across all cash flows, is used as the discount rate. All financing cash flows will be discounted at an after-tax cost of debt.

We can summarise which discount rates we should use to evaluate investment and financing decisions as follows:

Investment decisions	Financing decisions
Assets WACC	**Equity**
	Cost of equity K_e
	Debt
	After-tax cost of debt $K_d(1-t)$

Calculating the net present cost

Page 15-14 As an illustration of the lease-versus-borrow evaluation procedure we will consider Example 15.1.

Example 15.1: Calculating the net present cost of borrowing and leasing

Master Manufacturing is considering the purchase of plant for R5m. A loan could be obtained at an effective rate of 9.58904% repayable annually in arrears over five years, to cover the purchase price of the machine. Alternatively, the machine could be leased on a five-year contract for R1.25m per year, payable annually in advance. If the machine is owned, service and maintenance charges will be R250 000 per annum, whereas the lease charge includes maintenance and servicing. The salvage value of the machine in five years' time is expected to be R700 000. In terms of Section 12C, the company is allowed a tax write-off of 40% of cost in Year 1 and 20% of cost per year in years 2 to 4.

Assume a company tax rate of 27%. The company has a before-tax cost of debt of 9.58904%and a weighted-average cost of capital of 12%.

The present cost of leasing

The net present cost (NPC) of Master's leasing proposal is shown in Table 15.1. Cash outflows are shown as negative cash flows.

Notice that the lease payments and their relative tax allowance are not perfectly synchronised because the lease agreement calls for the lease payment to be made at the start of each year while the tax shield would be realised only at the year-end. The timing of these lease payments shown in Table 15.1 is based on the present-value convention of expressing cash flows on an end-of-the-year basis. As a result, the first lease payment made at the beginning of year one is shown as occurring at period zero. The discount rate used was based on the after-tax cost of debt of 7.0% [9.58904% × (1 – 0.27)].

Table 15.1 The net present cost of Master's leasing alternative

Year	0	1	2	3	4	5
Lease payment	-1,250,000	-1,250,000	-1,250,000	-1,250,000	-1,250,000	
Tax shield	27%	337,500	337,500	337,500	337,500	337,500
Cash Flows	-1,250,000	-912,500	-912,500	-912,500	-912,500	337,500
PV factor *[No rounding]*	$1/(1+r)^t$ 1.00000	0.93458	0.87344	0.81630	0.76290	0.71299
	-1,250,000	-852,804	-797,013	-744,872	-696,142	240,633
Net Present Cost	7.0%	After-tax rate = rate x (1-		9.58904%	0.73	7.0%
	-4,100,197	t)				

The present cost of borrowing and buying

As an alternative to leasing the asset, Master could borrow and buy. The first step in the determination of the present cost of borrowing is the calculation of the equal annual repayments required on a loan. This is calculated by taking the principal sum that is borrowed and dividing it by the present-value annuity factor for 9.58904%, for five years.

We cannot use Table D in this case as the closest PV factor is for 10%, which would result in an instalment of R1 318 983 [R5m/3.7908] per year for 5 years. We can, however, use the following formula to determine that the PV factor should be 3.830923:

$$PVA = PMT\left[\frac{1-\frac{1}{(1+r)^n}}{r}\right]$$

$$(1-(1/(1+0.0958904)^5))/0.0958904 = 3.830923$$

Page 15-15 The annual instalment is determined as follows:

$$\text{Instalment} = \text{principal/annuity factor} = 5\,000\,000/3.830923 = R1\,305\,169$$

Alternatively, we can use a financial calculator to determine the annual instalment as follows:

N	I/YR	PV	PMT	FV
5	9.58904%	5,000,000	-1,305,169	0

Table 15.2 then goes further by splitting this annual payment into tax-deductible interest and the non-deductible repayment of capital.

Table 15.2 Loan amortisation schedule

Year	Beg.Bal	Payment	Interest	Principal	End. Bal.
1	5,000,000	1,305,169	479,452	825,717	4,174,283
2	4,174,283	1,305,169	400,274	904,895	3,269,389
3	3,269,389	1,305,169	313,503	991,666	2,277,723
4	2,277,723	1,305,169	218,412	1,086,757	1,190,966
5	1,190,966	1,305,169	114,202	1,190,966	0
Total		6,525,843	1,525,843	5,000,000	

Having calculated the loan amortisation schedule, it is then necessary to discount the cash flows to obtain a present cost associated with the loan payments. This calculation is displayed in Table 15.3.

Table 15.3 The present cost of the loan

Year		0	1	2	3	4	5
Loan repayment			-1,305,169	-1,305,169	-1,305,169	-1,305,169	-1,305,169
Tax shield (interest × t)	27%		129,452	108,074	84,646	58,971	30,835
		0	-1,175,717	-1,197,095	-1,220,523	-1,246,197	-1,274,334
PV Factor	$1/(1+r)^t$	1.00000	0.93458	0.87344	0.81630	0.76290	0.71299
		0	-1,098,800	-1,045,589	-996,310	-950,718	-908,582
Net Present cost at	7.0%	**-5,000,000**					

Note: tax shield in Yr1: 479 452 x 27% = 129 452, Yr2: 400 274 × 27% = 108 074

It may be surprising that the present value of these cash flows is the same as the original capital sum. However, this is quite logical. Effectively we are taking a loan of R5m and adding to it interest of 7% based on repayments over five years, and then discounting these cash flows back at 7% to arrive at the original capital sum. Note that the interest is calculated at a before-tax cost, but as an allowance is made for the interest tax shield, the effective rate is really the aftertax cost.

If this is indeed the case, there is no need to calculate the monthly loan instalment, the loan amortisation schedule, or the present value of the loan amortisation. In its place we can simply insert the capital sum. However, while this would occur in most of the cases, there are situations when the monthly loan instalment, the loan amortisation schedule, and the present value of the loan amortisation will need to be calculated. Such a situation will occur when the firm has an assessed loss or when the interest rate that is applicable to a particular loan is not the same as the discount rate which is the after-tax cost of debt. Page 15-16

In Table 15.4 the present cost of Master's borrowing alternative is shown:

Table 15.4 The net present cost of Master's borrowing alternative

Year		0	1	2	3	4	5
PV of Loan		-5,000,000					
Depreciation tax shield			540,000	270,000	270,000	270,000	
Residual value							700,000
Recoupment tax							-189,000
Maintenance			-250,000	-250,000	-250,000	-250,000	-250,000
Tax shield maintenance	27%		67,500	67,500	67,500	67,500	67,500
		-5,000,000	357,500	87,500	87,500	87,500	328,500
PV Factor	$1/(1+r)^t$	1.00000	0.93458	0.87344	0.81630	0.76290	0.71299
		-5,000,000	334,112	76,426	71,426	66,753	234,216
Net Present cost at	7.0%	**-4,217,067**					

Note: Depreciation deduction is 40% of cost in year 1, 20% per year in years 2-4. Tax shield = deduction × tax rate.

The principal sum is shown in this evaluation as a cash outflow at time zero, as it is equal to the present value of the loan repayments and interest shields. Other cash flows which need to be considered are as follows:

- the depreciation deduction is 40% of cost in Year 1 and 20% of cost per year in Years 2 to 4. The tax saving in Year 1 will be R540 000 (40%×R5m × 27%). The tax saving in Years 2 to 4 will amount to R270 000 each year (20% × R5m × 27%);
- the maintenance charge of R250 000 per annum (after-tax R182 500) which must be paid if the asset is purchased; and ■ the recoupment at the end of Year 5 will result in a tax cash outflow of R189 000 (R700 000 × 27%). This is because the tax value then will be zero and the recoupment will amount to R700 000 (R700 000 – 0 = R700 000).

Again, these cash flows are all discounted at the after-tax cost of debt.

From the analysis in Table 15.1 and the analysis in Table 15.4 it can be seen that it would be more cost effective for Master to lease the plant as the net present cost would be R4 100 197 compared to a net present cost of R4 217 067 if Master were to borrow and buy. In such an instance Master should lease. The net advantage of leasing (NAL) is R116 869 (4 217 067 – 4 100 197).[1]

We could undertake the leasing evaluation on an incremental basis to reach the same conclusion and net advantage of leasing (NAL).

Table 15.5 The net advantage of leasing on an incremental basis

Year		0	1	2	3	4	5
Lease payment		-1,250,000	-1,250,000	-1,250,000	-1,250,000	-1,250,000	
Tax shield			337,500	337,500	337,500	337,500	337,500
Proceeds – cost		5,000,000					
Depreciation tax shield foregone		-540,000	-270,000	-270,000	-270,000	0	
Residual value foregone							-700,000
Recoupment tax avoided							189,000
Net maintenance avoided			182,500	182,500	182,500	182,500	182,500
		3,750,000	-1,270,000	-1,000,000	-1,000,000	-1,000,000	9,000
PV Factor	$1/(1+r)^t$	1.00000	0.93458	0.87344	0.81630	0.76290	0.71299
		3,750,000	-1,186,916	-873,439	-816,298	-762,895	6,417
NAL	7.0%	**116,869**					

The net advantage of leasing and NPV

 So far we have assumed that it is necessary to keep the investment and the financing decisions separate. This distinction enabled us to have a separate capital budgeting model and a separate financing model which used different discount rates. However, there could be instances where it is necessary to evaluate the investment decision in the light of low cost forms of asset-specific finance. It must be stressed that such a situation will occur only when the finance is inextricably linked to the acquisition of a particular asset, the financing opportunity being lost if the asset is not acquired. If the asset comes with favourable asset-specific financing, then there is an effective integration of the investment and financing decisions. Why? The low cost assetspecific financing is dependent on the firm investing in a particular asset. This means that the manufacturer of equipment will offer directly or indirectly lease financing at an implicit cost that is less than the firm's normal cost of debt. The manufacturer is effectively reducing the cost of the equipment without reducing prices. How the investment and financing decisions relate is shown in Figure 15.4.

The starting point is always to assess whether the net present value is positive. In the capital budgeting section we learnt that where the net present value is positive, we would invest in the project and where it is negative, we would not. If the net present value is positive, Figure 15.4 indicates that the next stage is to assess whether the net advantage of leasing is positive. The net advantage of leasing is the difference between the present value of the lease costs and the present value of borrowing and purchasing. This was examined in Example 15.1. If the net advantage of leasing is positive, we would lease the asset, while if it is negative, we would borrow and purchase. Note that as the net present value is positive, we will invest in the project and it is only the method of financing that is debatable.

If we look at the right-hand side of Figure 15.4, we examine a situation where the net present value is negative. However, before discarding the project out of hand, it is necessary to calculate the net advantage of the asset-specific lease. If the net advantage of leasing is negative, we would ignore the project. If the net advantage of leasing is positive, we need to ascertain whether the net present value and net advantage of the asset-specific lease, when added together, are positive. This will indicate whether the positive net advantage of leasing is large enough to offset the negative net present value. If it is not, we would again ignore the project. However, if the total is positive, we would lease the asset and go ahead with the project.

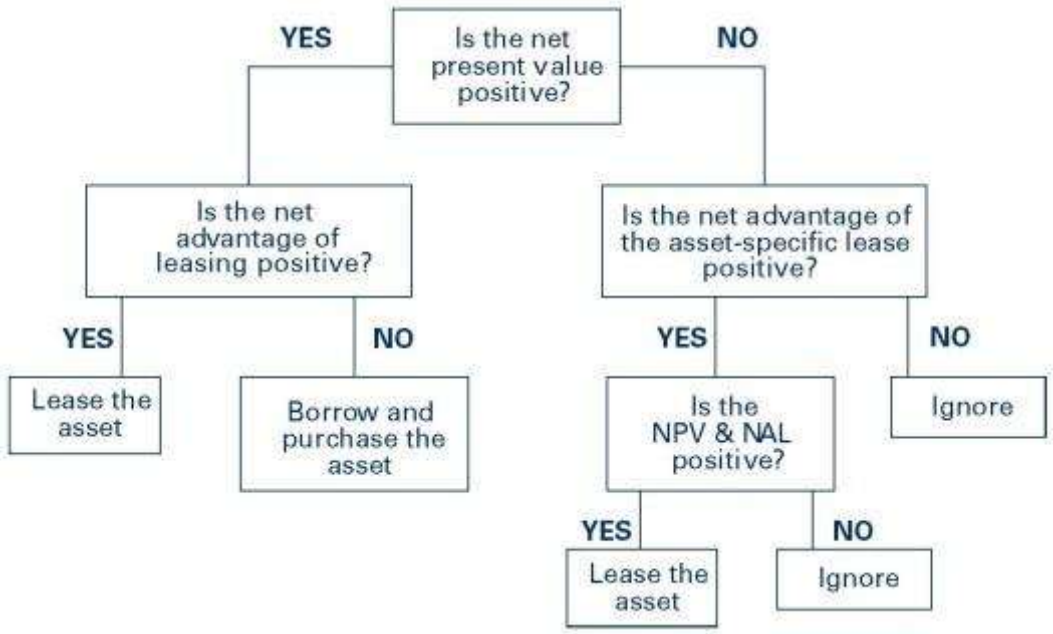

Figure 15.4 Combining the investment and financing decision where there is asset-specific finance

If we calculate the net advantage of leasing for Master Manufacturing we see that it comes to an amount of R116 869. This is the difference between the net present cost of borrowing and purchasing and the net present cost of leasing. Page 15-18

Assuming that the leasing opportunity is asset-specific, Master would go ahead and lease the asset as long as the original capital budgeting net present value was not worse than a negative R116 869. This is the lowest net present value that, when combined with the net advantage of leasing, will not yield a combined negative total present value.

The asset-specific finance could also be in the form of loan finance. Therefore, although we are currently evaluating leasing, it would mean that we would need to take into account the benefit of any low cost asset specific financing in the investment decision. In the luxury car market, manufacturers will offer special financing deals with the acquisition of a vehicle which effectively reduces the cost to customers without the manufacturer reducing prices.

FROM THE REAL WORLD ...

In FY2023, **Pick n Pay** reported net lease interest of R1 254m but net funding interest of only R171m. Depreciation on right-of-use assets was R2 148m as compared to depreciation of R1 320m on property, plant and equipment. Right-of-use assets were R11 195m and lease liabilities were R17 604m. Yet, Pick n Pay's total equity was only R3 703m. This indicates the significant impact that IFRS16 has had on Pick n Pay's financial statements.

MTN's sale and leaseback of its cell phone towers

In September 2014, MTN announced that it was selling its 9 200 towers in Nigeria to IHS, who would take over operational control and management of the towers and would lease back the towers to MTN. In 2013, MTN sold its 911 towers in Côte d'Ivoire to IHS for US$141m and its 820 towers in Cameroon for US$143m. IHS would manage the towers, the sites and the related passive infrastructure. MTN stated in its 2013 annual report that, "the transactions resulted in sale and leaseback transactions classified as operating leases". In 2012, MTN entered into a sale and leaseback arrangement with American Tower Corporation (ATC) for 962 of MTN's towers and sites in Uganda at a purchase price of US$175m. In this transaction, ATC owns 51% of the tower holding company and 49% is owned by MTN. Again, the transaction resulted in a sale and leaseback transaction, which was classified as an operating lease.

Why? How does it work? Why would MTN wish to sell its towers, which are surely a key component of its network?

Let's try and understand what ATC does. ATC owns and operates the real estate necessary for wireless communications networks. A sale of the towers will include the tower structure, which will be constructed mainly of galvanised steel with space for a number of tenants to place their communication equipment on each tower. ATC will also own the land, which will be subject to a long-term lease. MTN will continue to own and maintain the antenna and microwave equipment, as well as other base station equipment and the coaxial cable. ATC's business model involves buying towers and having more than one tenant sharing each tower and land to place antennae and other equipment as this generates additional revenue and there are few incremental costs Prior to the sale of its towers, only the MTN equipment was on each MTN owned tower. Sharing towers with other carriers will reduce operating costs over time and towers can be put to other uses, such as hosting television and radio transmitters. Furthermore, carriers can reduce the time to get to market by leasing space rather than undertaking the acquisition of locations, zoning and building the passive infrastructure. Also, carriers such as MTN need to continue to invest in 3G and 4G networks and related capital expenditure as the demand for data continues to grow. Voice is mainly carried over 2G and 3G networks while 4G/LTE networks mainly carry data.

ATC uses an illustrative example to set out the economic incentive for mobile companies (carriers) such as MTN to sell their towers and enter into lease arrangements. The following represents the present value of the purchase (build) and ownership option relative to the leasing alternative. The Weighted Average Cost of Capital (WACC) employed is assumed to be 9%, the cost of the site is $225 000, and operating costs amount $1250 per month with an escalation factor of 3% per year. The tower lease alternative represents a monthly lease payment of $1800.

Term (years)	Carrier Build	Tower Lease	Savings

10	$333,079	$158,720	$174,359
20	$394,433	$253,293	$141,140

The move to data requires continued investment in sites, infrastructure and equipment. Furthermore, there may be tax advantages relating to this structure. Political risk may also be reduced as there are now two major companies involved in the ownership of wireless network assets. Leasing wins for strategic reasons and it wins in terms of the numbers.

MTN's leases were classified as operating leases but the introduction of IFRS 16 from 2019 meant that the group needed to include a right-of-use asset and the present value of future lease payments as a liability. MTN had a lease expense of R13.4bn in 2017 and operating lease commitments of R95bn (undiscounted) which were not included in the Statement of Financial Position. MTN stated the following in its 2017 annual financial statements:

> On adoption of IFRS 16 operating lease costs will no longer be recognised as operating expenses. The extent of the reduction in lease expenses is dependent on the application of the practical expedients in IFRS 16 regarding the separation of lease and non-lease components and the impact of the application of the low-value asset exemption. The new standard will require the recognition of lease liabilities and corresponding right- of-use assets. The group will recognise depreciation on the right-of-use assets and interest on the lease liabilities over the lease term in profit or loss. The initial lease liabilities and right-of-use assets recognised upon transition to IFRS 16 would likely be representative of the non-cancellable lease commitments, discounted at an appropriate rate as applicable to the operation in which the lease arises, after taking into account the impact of the practical expedients and transitional elections applied by the group. It is anticipated that while the EBITDA and the related EBITDA margin will improve significantly, depreciation and finance charges will also increase significantly. Due to the impact of reducing finance charges over the life of the lease, the impact on earnings will initially be dilutive, before being accretive in later periods.

MTN reported the following in 2017 with regard to IAS 17 but we think that this may have some relevance to IFRS 16 in relation to its tower leases.

> The requirements to be met in order to conclude that an arrangement constitutes or contains a lease are as follows:
>
> - The provision of a service in terms of the arrangement should be dependent on the use of one or more specific assets.
> - The arrangement must convey a right to use these assets.
>
> All other arrangements that do not constitute or contain leases are treated as service level agreements; the costs are expensed as incurred. The group has resolved that an arrangement contains a lease as defined in IAS 17 Leases where the arrangement provides an exclusive right to use specific tower space which is more than an insignificant part of the tower asset.

In 2021, MTN sold about 5700 tower sites and power assets to IHS and has leased back space on the towers. Tower spaces have been leased back to MTN for 10 years with an option to renew for another 10 years. This part of the transaction has been reported in terms of IFRS16. The transfer of MTN's power assets to IHS means that IHS will supply MTN with electricity. IFRS16 will not apply to this part of the transaction. MTN states:

> As MTN SA is transferring its power assets and will be receiving electricity and other services going forward, the Group accounted for this part of the transaction as a disposal of property, plant and equipment as it no longer has the right to control the use of an identifiable asset. The electricity utility and power backup service arrangement are accounted for as a service arrangement and recognised as an expense as the service is received.

In FY2022, MTN reported right-of-use assets of R50.625bn and lease liabilities of R58.344bn. Lease liability interest expense amounted to R6.792bn. Page 15-20

6 THE ADJUSTED PRESENT VALUE APPROACH

Normally the decision to invest in a capital project will impact on other financial decisions made by the company and these side-effects have to be taken into account when a project is evaluated.

Probably the most common impact is the effect that the investment decision has on a company's borrowing capacity. For example, if a company has a 50% target debt ratio, it will limit debt to 50% of the book value of its assets. However, if the company invests more it will borrow more in order to maintain its capital structure.

This increase in debt capacity yields increased interest tax shields. In Chapter 14, consistent with Modigliani and Miller's original theory, we arrived at the conclusion that the value of a firm was independent of its capital structure except for the present value of the interest tax shield. This gave us the following formula:

$$V_L = V_U + \text{PV of interest tax shield}$$

where: V_L = value of levered firm

V_U = value of unlevered firm

This theory supports the stance that the value of the firm should be calculated in two steps: firstly, using a base case under all equity financing and, secondly, adding the present value of the interest tax shield resulting from a departure from all equity financing. This, in essence, is an adjusted present value calculation for the firm as a whole.

This approach can be repeated for particular projects. For example, consider a project costing R100m which had an after-tax cash inflow of R27.5m for five years. If the project's allequity (unlevered) cost of capital was 12% based on the project's business risk, the net present value would be:

	0	1	2	3	4	5
Cost	(100,000,000)	27,500,000	27,500,000	27,500,000	27,500,000	27,500,000
Cash flows						
	(100,000,000)	27,500,000	27,500,000	27,500,000	27,500,000	27,500,000
NPV (12%)	(868,654)					

As the NPV is negative, the financial manager would be inclined to ignore the project.

However, the investment in this project would increase the company's assets by R100m and therefore, if the 50% target debt ratio is to be maintained, the company will need to borrow R50m more. Further, this capital structure will be maintained if we assume that the depreciation is in line with the capital repayment. If we assume that this R50m loan is repaid in equal instalments, the amount to be repaid annually, if an interest rate of 9% is applicable, would amount to R12 854 623 (R50m/3.8896513).[2] The value of the interest tax shields, assuming a 27% company tax rate, is shown in Table 15.6. This is the value of the additional debt capacity contributed to the company by the project. Page 15-21

Table 15.6 Calculations of the present value of the interest tax shield

Year	Beg.Bal a	Payment b	Interest c	Principal d = b-c	End. Bal. e = a-d	Interest tax shield f = c × tax rate	PVFactor 1/(1+r)t g	PV tax shield h = f × g
1	50,000,000	12,854,623	4,500,000	8,354,623	41,645,377	1,215,000	0.9174	1,114,679
2	41,645,377	12,854,623	3,748,084	9,106,539	32,538,838	1,011,983	0.8417	851,766
3	32,538,838	12,854,623	2,928,495	9,926,127	22,612,711	790,694	0.7722	610,561
4	22,612,711	12,854,623	2,035,144	10,819,479	11,793,232	549,489	0.7084	389,272
5	11,793,232	12,854,623	1,061,391	11,793,232	0	286,576	0.6499	186,254
Total		64,273,114	14,273,114	50,000,000		3,853,741		3,152,531

We obtain the adjusted present value by adding this amount to the project's existing present value:

$$APV = NPV + PV \text{ interest shield}$$

where: NPV = base case NPV at the cost of equity

$$APV = (R868\,654) + R3\,152\,531$$
$$= R2\,283\,877$$

While the net present value was negative, the adjusted net present value is positive, making the project worthwhile.

How does the APV approach reconcile with the NPV calculated by using the traditional weighted average cost of capital? If the NPV is now R2 283 877, the PV of the cash flows must be equal to R102 283 877 [NPV = PV – cost]. If we assume that the PV of R102 283 877 is the initial outlay, then we can use the IRR methodology to determine the effective discount rate as follows:

	0	1	2	3	4	4
PV & cash flows	- 102,283,877	27,500,000	27,500,000	27,500,000	27,500,000	27,500,000
IRR	**10.7483%**					

The effective discount rate is found to be 10.7483%, which is the implicit weighted average cost of capital. This rate can be compared to the weighted-average cost of capital calculated in the traditional manner.

In this example, the all equity cost of capital is given as 12% while the after-tax cost of debt is 6.57% (9% × (1 – 0.27)). However, to calculate the weighted average cost of capital, we need to estimate the levered cost of equity. Consistent with the Modigliani and Miller theory, we would expect the before tax cost of capital to remain at 12% irrespective of the amount of debt. Thus with a debt ratio of 50%, the following would hold:

	Cost	Weight	Overall
Average before-tax cost of capital			12.0%
Before-tax cost of debt	9%	50%	4.5%
Before-tax cost of equity	×%	50%	7.5%

The levered cost of equity is 15% (7.5%/0.50) and the weighted average cost of capital would be calculated as:

	Cost	Weight	Overall
Cost of Equity	15.00%	50%	7.500%
After-tax cost of debt	6.57%	50%	3.285%
Weighted Average Cost of Capital (WACC)			**10.785%**

 The APV and traditional NPV analysis will give similar results. The APV's implicit discount rate is 10.7483% as against 10.785% for the traditional NPV. The question that remains is: when is it appropriate to use an APV rather than a traditional NPV?

The APV approach is best suited to a situation where it is difficult to estimate the weighted average cost of capital possibly as a result of the project or investment

having a significantly different risk profile to that of the firm as a whole.

SUMMARY

The accounting for leases in terms of IFRS 16 requires that a company recognise a right-of- use asset and recognise a lease liability which is the present value of future lease payments in terms of non-cancellable lease commitments. A contract will be identified as a lease if there is an identified asset and the lessee obtains the economic benefits and directs its use. Leases can be categorised into *direct leasing, sale and lease-back*, and *leveraged leasing*.

Leases can be evaluated by cash flow analysis. We start with the assumption that an asset, as a result of a positive net present value, will be acquired, and that the acquisition will be financed either by debt or a lease. Next we develop the annual net cash outflows associated with each financing plan. Finally we discount the two sets of outflows at the company's aftertax cost of debt and choose the alternative with the lowest net present cost. The difference between the net present cost of leasing and borrowing to purchase is called the *net advantage of leasing.*

The *adjusted present value* approach suggests a base case NPV should be calculated using the project's all-equity cost of capital. This NPV is combined with the present value of the income tax shield to arrive at the APV.

Although the initial growth in leasing was mainly due to tax advantages, leasing also offers many other advantages such as flexibility, standardisation, avoiding technological obsolescence, a possible reduction in operating leverage, and reduced restrictions on borrowing.

SELF-STUDY PROBLEMS

S15.1

Cross-State Transport Ltd is involved in transporting commodities from Durban to Johannesburg. The company is required to replace its trucks regularly owing to the importance of providing a reliable service to its customers. The company is considering investing in a new truck at a cost of R1.2m. Cross-State's bank has indicated that it can offer the company a lease with payments of R292 510, payable annually in arrears for five years. There is no residual payment or value. Ignore taxation.

Required:

(a) What is the interest rate implicit in the lease?

(b) Assume that the bank wishes to keep the lease payment at R292 510 but requires that the annual payments are payable in advance. How does this change the interest rate implicit in the lease?

(c) Assume that the truck has an expected residual value of R450 000 at the end of five years and the bank takes this residual value into account in setting the lease payments, which are payable annually in advance. How would this change your lease payment?

S15.2

Just-in-Time Airlines (JIT) needs to invest in a 10-year-old Airbus plane at a cost of R200m. Air Finance Corporation specialises in the financing of planes and has offered the company either a 5-year lease requiring annual payments of R40m per year, payable in advance or a Page 15-23 loan which is repayable in a single repayment at the end of 5 years. The leasing option includes maintenance and this is worth R4m per year, which would have been payable in advance. The tax rate is 27% and you can assume that the tax benefits and costs are realised one year after the cash flows occur, except for the tax effects of the residual payment or value which will occur in Year 5. If the company takes the loan option, the company will be required to pay for ongoing maintenance of R4m per year payable in advance. The financing company has agreed to a residual payment of R40m in 5 years' time if JIT wishes to purchase the plane at the end of the lease. Yet the expected residual value in 5 years' time is expected to be R80m. The cost of capital is 12% and the before-tax cost of debt is 8.219%. The company is able to depreciate the cost of the plane on a straight-line basis over 5 years. What is the net advantage of leasing the plane?

S15.3

MedQ Ltd manufactures medical devices and wishes to improve the quality of one of its major product lines, the BPQ monitor, which accurately measures a patient's blood pressure at home. The company wishes to evaluate the financing of a new manufacturing machine.

A new machine can be purchased for R150m in order to replace the existing machine. The new machine has an estimated useful life of four years at the end of which the salvage value is expected to be R40m.

In terms of section 12C, the new machine would be subject to a 40% depreciation deduction in Year 1 and a straight-line 20% depreciation deduction per year in Years 2 to 4. The company has built up an assessed tax loss of R20m at the current date.

Profits from the production of other products are expected to result in taxable income of R130m per annum for the next four years. The BPQ monitor is expected to produce an additional taxable income of R80m before allowing for financing tax shields.

The new plant can be financed by way of a loan bearing interest at 9.589%, being the normal before-tax cost of debt. The loan would be repaid in two equal instalments at the end of the second and fourth years, while interest is payable annually in arrears on the amount outstanding. Alternatively, a lease is available with instalments of R33m payable annually in advance for four years. At the end of the lease, the lessor will retain ownership of the machine. A company tax rate of 27% is applicable.

Required:

Advise management as to the most appropriate financing alternative assuming it decides to invest in the new machinery.

Solutions to self-study problems

S15.1 Solution

1. If the lease payment is payable annually in arrears, the interest rate (IRR), that will ensure that the present value of future lease payments is equal to the cost, is 6.98%. This is the interest rate implicit in a lease payment of R292 510 per year for 5 years.

	0	1	2	3	4	5
Cost	1,200,000	-292,510	-292,510	-292,510	-292,510	-292,510
Lease payments						
	1,200,000	-292,510	-292,510	-292,510	-292,510	-292,510
IRR	**6.98%**					

You can use a financial calculator, by doing the following steps: Page 15-24

N	I/YR	PV	PMT	FV
5	**6.98%**	**1,200,000**	-292,510	0

Ensure that the calculator is set for annual payments in arrears by using the BEG/END function so that in the above case, BEGIN does not show up on your screen. The default is that payments are in arrears.

2. The bank may indicate that it will set the payment at R292 510 per year, but that this payment is payable annually in advance. Although the payment remains the same, the interest rate may be very different. Let's first use the financial calculator and reset it to BEGIN, implying that the payments are payable annually in advance.

N	I/YR	PV	PMT	FV
5	**11.00%**	**1,200,000**	-292,510	0

Using Excel and the IRR function, the cash flows now look as follows:

	0	1	2	3	4	5
Cost						
	1,200,000					
Lease payments	-292,510	-292,510	-292,510	-292,510	-292,510	
	907,490	-292,510	-292,510	-292,510	-292,510	
IRR	**11.00%**					

The decision of whether payments are in advance or in arrears can have a large impact on the interest rate that you are paying. This is particularly true for leases that require annual payments. The interest rate has increased from 6.98% to 11%.

3. The lease payment will be based on the cost of R1.2m less the present value of the residual value of R450 000 at the end of Year 5. The net amount that needs to be repaid to the leasing company is as follows:

	Amount	PV factor	Present Value
Cost	1,200,000	1.0000	1,200,000
Residual value	-450,000	0.5935	-267,075
Present value to be recovered from lease payments			932,925

The present value to be repaid in 5 instalments is R932 925 and as the first payment occurs today, we are only discounting 4 lease payments and the value of the first instalment is not discounted (the PV factor is equal to 1). What we are really doing is valuing an annuity in arrears for 4 years plus the value of the first instalment today. If you look at the Table D, the PV factor of 4 years for 11% is 3.1024. If we add 1 to this for the first instalment, we get to a PV factor of 4.1024.

	PV	PVFA	Lease Pmt
Determination of Lease Pmt	932,925	4.1024	227,410
[PV/(PV factor of an annuity for (n-1) +1)]			

Using a financial calculator, we find that after entering all the other numbers and then entering PMT, the result should be R227 412. There is a small rounding difference owing to the use of PV factors to four decimal places.

N	I/YR	PV	PMT	FV
5	**11.00%**	**1,200,000**	**-227,412**	- 450,000

S15.2 Solution

Page 15-25 The residual value of R80m is not included in the following analysis because the company will receive the residual value whether or not the company selects the lease option or the borrow- and-purchase option. As the NAL is positive, the company is better off by using a lease to finance the acquisition of the plane.

Net Advantage of Leasing (NAL)	R000	0	1	2	3	4	5
PV of Loan avoided (cost of asset)		200,000					
Lease payment		-40,000	-40,000	-40,000	-40,000	-40,000	
Tax saving of lease payment	1		10,800	10,800	10,800	10,800	10,800
Value of maintenance expense		4,000	4,000	4,000	4,000	4,000	
Tax saving on maintenance foregone	2		-1,080	-1,080	-1,080	-1,080	-1,080
Depreciation tax saving foregone	3		-10,800	-10,800	-10,800	-10,800	-10,800
Residual payment							-40,000
Tax [residual value-residual pmt]	4						-10,800
Tax on recoupment avoided	5						21,600
		164,000	-37,080	-37,080	-37,080	-37,080	-30,280
PV factor		1.0000	0.9434	0.8900	0.8396	0.7921	0.7473
PV of cash flows at loan interest rate		164,000	-34,981	-33,001	-31,132	-29,371	-22,628
Net Advantage of Leasing (NAL)		**12,886**	[As the NAL	is positive, the firm should lease]			

1. Lease Pmt x tax rate
2. Maintenance x tax rate
3. (Dost x Depreciation rate x tax rate
4. (Residual Value - Residual payment) x tax rate [The difference between residual value and residual payment is taxable]
5. (Residual value - tax value) x tax rate. The tax value would be zero at end of Yr5. [Recoupment: R80m - 0 = R80m] The borrow & purchase option would have resulted in a recoupment in Yr5. A benefit of leasing is that it avoids this.
6. Leasing and the borrow and purchase options both result in residual values of R80m and so these offset each other and are not included here.

S15.3 Solution

Financing decision: workings

1. Net present cost of leasing: 7% (9.589% × (1 – 0.27))

Leasing	Rm	0	1	2	3	4
Lease payment		-33.00	-33.00	-33.00	-33.00	
Lease tax shield (lease pmt × tax rate)			8.91	8.91	8.91	8.91
		-33.00	-24.09	-24.09	-24.09	8.91
PV Factor		1.0000	0.9346	0.8734	0.8163	0.7629
		-33.000	-22.514	-21.041	-19.665	6.797

Leasing	Rm	0	1	2	3	4
Net Present Cost		-89.422				

2. Net present cost of borrowing: 7%

Borrowing	Rm	0	1	2	3	4
PV of Loan		-150.00				
Depreciation tax saving			16.20	8.10	8.10	8.10
Residual value						40.00
Recoupment						-10.80
		-150.00	16.20	8.10	8.10	37.30
PV Factor		1.0000	0.9346	0.8734	0.8163	0.7629
		-150.000	15.140	7.075	6.612	28.456
Net Present Cost		**-92.717**	[Note: Depreciation tax saving = Cost × Depreciation % × tax rate]			

The net advantage of leasing is R3.295m (R92.717m – R89.422m). It is important that we include a realistic estimate of the residual value in computing the net present cost of the borrowing and purchase option. For example, let's assume that although a realistic residual value is R40m, the company has included only an estimated value of R10m. The net present cost of the Page 15-26 borrowing and purchase option rises dramatically to over R109.424m whilst the NPC of leasing remains at R89.422m. This leads the company to select the leasing option, yet the real difference between the financing options in this case is marginal. The assessed loss is not relevant as earnings from other products will fully utilise the tax loss in Year 1.

Borrowing	Rm	0	1	2	3	4
PV of Loan		-150.00				
Depreciation tax saving			16.20	8.10	8.10	8.10
Residual value						10.00
Recoupment						-2.70
		-150.00	16.20	8.10	8.10	15.40
PV Factor		1.0000	0.9346	0.8734	0.8163	0.7629
		-150.000	15.140	7.075	6.612	11.749
Net Present Cost		**-109.424**	[Note: Depreciation tax saving = Cost × Depreciation % × tax rate]			
Workings		Rm				
Cost		150.00				
Depreciation - Yrl		-60.00	[40% × 150m]			
Depreciation - Yrs2-4		-90.00	[20% × 150m×3]			
Tax value - end of Yr4		0.00				
Residual value		10.00				
Recoupment		**10.00**				
Tax		2.70				

Let's go back to an estimated market value of R40m for the equipment at the end of Year 4. If the company decided to retain the equipment and there was an option by the company to purchase the equipment from the lessor at the market value at the time, then the analysis becomes more complex. Let's assume that the company intends to use the equipment for a further 5 years after the end of Year 4.

We would now need to take into account the fact that the leasing option will have an outflow at the end of Year 4, but this would be subject to a depreciation deduction of 20% per year straight-line. Assume that the value of the equipment at the end of Year 9 is expected to be zero. The net present costs of the lease and borrow and purchase options are as follows:

Leasing	0	1	2	3	4
Lease payment	-33.000	-33.000	-33.000	-33.000	
Residual value					-40.000
PV of depreciation tax shields Yr 5-9					8.856
Lease tax shield		8.910	8.910	8.910	8.910
	-33.000	-24.090	-24.090	-24.090	-22.234
PV Factor	1.0000	0.9346	0.8734	0.8163	0.7629
	-33.000	-22.514	-21.041	-19.665	-16.962
Net Present Cost	**-113.1817**				
PV of depreciation tax shields					
Depreciation	40.00	20%	8.000		Alternatively
Depreciation tax shields	8.00	27%	2.160	=PV(0 07,5,-2.16,0,0)	
PV of depreciation tax shields	2.16	4.1002	8.856		8.856 Page 15-27

Borrowing	0	1	2	3	4
PV of Loan	-150.00				
Depreciation tax saving		16.20	8.10	8.10	8.10
Residual value (Zero in Yr 9)					
Recoupment					
	-150.00	16.20	8.10	8.10	8.10
PV Factor	1.0000	0.9346	0.8734	0.8163	0.7629
	-150.000	15.140	7.075	6.612	6.179
Net Present Cost	**-114.9935**				

The net present cost of leasing remains the preferred option. However, the net advantage of leasing has fallen from R3.295m to R1.8118m (R114.9935m – R113.1817m).

From this analysis it would appear that the lease option is preferable as it has a lower net present cost.

APPENDIX 15.1: OPERATING LEASE CAPITALISATION AND OTHER PERTINENT LEASING ISSUES

by Johnathan Dillon M.Com CA(SA)

Leasing is clearly a significant activity that is increasing for many entities, both globally and within our local economy, and needs to be accounted for in terms of its economic reality.

Due to the move towards bringing operating leases "onto the balance sheet" of lessees, the dissertation that I submitted in partial fulfilment of my M.Com (Finance) degree was entitled "*Impact of constructive capitalisation of operating leases on South African companies considering new proposed lease accounting rules*".

Industry analysis

From research performed in the US, UK, Germany, Canada, Belgium and Netherlands, it was found that the following industries made substantial use of operating leases (that is, off-balance sheet debt): retailers, restaurants, hotels, transportation and industrial products. My research therefore focused on JSE-listed companies, as at 9 September 2013, that operated within the same sectors (namely, food and drug retailers, general industrials, general retailers, industrial transportation and travel and leisure) and it confirmed that operating lease usage was widespread.

Operating versus finance lease usage

While 100% of the companies included in my study had entered into operating leases, it was found that only 47.9% of the same companies had recognised liabilities in respect of finance leases. Furthermore, when the disclosed future minimum operating lease payments were capitalised, as described in the next section, it was found that the present value of the operating lease payments (unrecognised operating lease liability) was, on average, 33.7% of the total recognised debt for these companies. However, for the same companies, it was established that recognised finance lease liabilities comprised only 4.6% of the total recognised debt, on average.

The impact on key financial statement ratios when capitalising operating leases:

As expected, due to operating lease commitments being unrecognised debt, it was found that leverage and other debt-related ratios of South African companies are substantially affected when constructively capitalising operating lease commitments. Selected results are indicated in the table below: Page 15-28

Financial ratio	Mean precapitalisation (reported figures)	Mean postcapitalisation	Mean absolute change
Debt-to-equity (market value)	79.4%	92.5%	13.1%
Times interest earned	111.2	8.0	-103.6

Retailers were found to be the most affected industry when capitalising operating leases. For example, Pick n Pay Holdings Limited's debt-to-equity ratio, based on market value, increased from 97.7% to 176.4%. The debt ratio (total assets divided by total debt) is less affected by the capitalisation of operating leases due to the fact that a right-of-use asset is recognised in addition to the lease obligation relating to the future minimum operating lease commitments. However, based on straight-line amortisation of the right-of-use asset, it is important to note that an operating lease asset recognised is always less than the associated lease liability recognised at any point after inception of the lease (this is illustrated in the Economic Value Added example in the next section).

Summary

Operating lease usage in South Africa, considering the five industries analysed, is extensive. Furthermore, when operating leases are capitalised, the impact thereof on debt-related financial ratios is substantial. Therefore, the annual financial statements of many South African companies are going to look very different and some of their key financial ratios are going to change considerably.

The capitalisation of off-balance sheet operating leases is not a new phenomenon as it is one of the most commonly known adjustments made when calculating Economic Value Added and the capitalisation of all non-cancellable operating leases was taken into account in the revised ZETA® failure prediction model developed by Altman, Haldeman and Narayanan in 1977.

In conclusion, it is questionable whether leases will continue to be used as extensively as they currently are due to the loss of the off-balance sheet status of operating leases. Will the change in lease accounting affect the leasing industry negatively? This may not be the case if the other benefits of leasing an asset, as opposed to buying an asset, will result in companies leasing to the same extent. Only time will tell.

References:

Dillon, J, 2014. Impact of constructive capitalisation of operating leases on South African companies considering new proposed lease accounting rules.

The pervasive impact of operating lease capitalisation in relation to Economic Value Added and Free Cash Flow Valuations

An illustration of how to capitalise an operating lease using the before tax cost of debt and the impact thereof on the capital structure and the WACC calculation of an entity is included in Chapter 7 within the section entitled "Operating leases, capital structure and the weighted-average cost of capital" (Section 10). The inclusion of operating leases in the WACC calculation leads to a decrease in the company's WACC (as debt is cheaper than equity). However, a large portion of the decrease should be offset by an increase in the company's equity beta, increasing the cost of equity, due to the resulting higher debt/equity ratio (increased financial leverage). A further point to note is that you can correctly assume that the

market already recognises the operating lease obligations and that the company's beta therefore already factors in the implicit leverage (increased financial risk) arising from the use of operating leases, in which case no unlevering and relevering of the existing beta is required.

However, to further illustrate the pervasive impact of capitalising operating leases, the impact thereof on Economic Value Added (EVA) is considered. EVA is covered in Chapter 5 and is calculated as follows:

$$EVA = NOPAT - (Investment \times WACC)$$

Page 15-29 Considering the EVA formula, in addition to the changes to the WACC summarised above, NOPAT will change as a result of the following:

- The removal of the operating lease expense from NOPAT; and
- The inclusion of the asset write-off relating to the right-of-use lease asset recognised.

  Note: The interest expense relating to the lease liability is not adjusted for as NOPAT is operating profit before interest - the cost of debt is included in the WACC calculation.

The investment will also change as the present value of the capitalised operating leased assets will be added thereto. This is best illustrated by way of a simplified example. Assume the following information has been extracted from a listed company's latest annual report:

Total net operating assets (equity plus long-term debt) at year-end	R900m
Net operating profit after tax (NOPAT)	R112m
Operating lease payments incurred in the current year	R20m
Minimum lease payments due under operating lease agreements:	
Up to one year from year-end	R15m
Two to five years from year-end	R44m
Average useful life of company's operating assets	4 years
Cost of debt (before tax)	9% p.a.

Assume also that the company's WACC has been reliably estimated as 13% when excluding the impact of operating leases. Furthermore, if operating leases are capitalised (through an adjusted capital structure and beta) then the WACC equals 12%.

$$EVA \text{ (ignoring operating lease capitalisation)} = 112m - (900m \times 13\%) = -R5.0m$$
$$= R5.0m \text{ value destroyed}$$

In order to calculate the revised EVA figure, incorporating the impact of capitalised operating leases, the reported figures need to be adjusted and the following calculations are necessary in that respect [Note: these calculations incorporate a few simplified assumptions, including the fact that all cash flows occur at the end of every year]:

1. *Lease liability and asset value at the beginning of the current financial year = R61.0m (per analysis below)*

	Year	0	1	2	3	4	5	6
Operating lease payments (Rm)			-20	-15	-11	-11	-11	-11
		-	-20	-15	-11	-11	-11	-11
Net present value (using 9%)		-61.0						

Note: Year 0 is the start of the current financial year while Year 1 is the end of the current financial year or the beginning of the next financial year. The R11m operating lease payment in Years 3 to 6 is estimated as R44m/4 years = R11m per year, although this is more likely to be higher in Year 3 and then decline over the remaining three years as lease agreements come to an end.

1. *Depreciation charge for the year = 61.0 ÷ 4 = R15.25m*
2. *Interest charge for the year = 61.0 × 9% = R5.49m*
3. *Lease liability at year-end = 61.0 – 20 + 5.49 = R46.49m*
4. *Asset book value at year-end = 61.0 – 15.25 = R45.75m*

Revised NOPAT = 112m + [(20m – 15.25m) × 0.73] = 115.47m
Revised investment (total net operating assets) = 900m + 45.75m = 945.75m

Therefore, revised EVA (incorporating the capitalisation of operating leases):

EVA = 115.47m – (945.75m × 12%) = R1.98m

= R1.98m value created (which looks better than R5.0m value destroyed as initially calculated)

Page 15-30 With respect to valuations, especially the free cash flow valuation approach described in Chapter 6, operating leases should also be capitalised and adjusted for when performing a discounted free cash flow to the firm valuation of an entity. Similar adjustments as those described above would be applicable, such as the adjustment to the WACC. Furthermore, the operating lease expense must be removed from profit and the associated capital expenditure relating to the operating lease assets (calculated as the movement in the operating lease asset balance after adjusting for depreciation thereon) would be deducted in arriving at the free cash flow from operations. The present value of the operating lease liability must then be deducted from the value of the firm in order to calculate the value of equity. This approach is illustrated in Question 7.32.

QUESTIONS

Question 15.1

Leasing represented off-balance sheet financing prior to IFRS 16. Indicate how the introduction of IFRS 16 is likely to have affected a company's financial ratios such as debt-to-equity, the current ratio, interest cover and the EBITDA margin.

Question 15.2

Discuss the contention that leasing provides an opportunity to obtain debt financing in excess of a firm's debt capacity.

Question 15.3

What are the major advantages of leasing?

Question 15.4

Once it has been determined that an asset should be acquired, how should management decide whether the asset should be leased?

Question 15.5

Discuss the principle that financing cash flows should be discounted at the after-tax cost of debt rather than the weighted average marginal cost of capital.

Question 15.6

Trader Limited has decided to acquire additional warehouse space for storage purposes. The firm has located an existing warehouse which is suitable and this can be either leased or purchased. The firm can lease the warehouse for 10 years for an annual fee of R1.4m payable each year in advance. Alternatively, it can purchase the warehouse for R10m. In either case, all repairs and maintenance are the responsibility of Trader. The firm estimates that the residual value of the warehouse at the end of the ten years will be R12m. Assume a tax rate of 27%, that the cost of debt is 9.589%, and that the WACC is 12%. If purchased, the warehouse can be depreciated on a straight-line basis, over 20 years.

Required:

Should the warehouse be leased based on a present value analysis?

Question 15.7

Maxit Ltd is considering investing in a project that will result in annual before-tax cash flows of R48 million per year, for 5 years. The company will need to invest in equipment which the company can either lease or purchase.

Purchase: The cost of the equipment is R130 million and the expected residual value at the end of 5 years is zero. The depreciation deduction is 20% per year straight-line.

Lease: The Company is required to make annual lease payments of R32 million per year, payable in advance over 5 years. The tax deduction relating to the lease

payments will occur one year after each lease payment.

The tax rate is 27%. The company's cost of capital is 12%, whilst the after-tax cost of debt is 7%. What is the NPV of the project? If the NPV is positive, indicate whether the company should lease or purchase the equipment?

Question 15.8

XYZ has decided to invest in a computer-controlled sorting and mixing machine costing R12m. The machine has an economic life of seven years, after which no salvage value is expected. The company must determine whether it is better to finance the acquisition through debt or leasing. XYZ expects profits before taxes of R10m next year, R20m the following year, R30m the third year and R40m each year thereafter (before depreciation or leasing charges on the machine). It has a high degree of confidence in these estimates. The company has had break-even operations during the previous three years. The company will qualify for a depreciation deduction of 40% of cost in the first year and 20% in each of the subsequent 3 years.

If the machine is financed with debt, the National Bank is willing to extend a loan for the full purchase price at an interest rate of 9.589% payable in equal annual amounts over seven years. The bank also has a leasing division and has indicated that it is willing to provide lease finance for the acquisition of the machine over the seven-year period at R3m per annum, payable in advance. The company's cost of capital is 11%. The tax rate is 27%.

Required:

Should the company finance the acquisition of the device by means of the lease or the loan?

Question 15.9

Cape Trucking (Pty) Ltd transports commodities and food products from Cape Town to Johannesburg and Bloemfontein. The company has decided to invest in a new truck which will cost R1.8 million. The company is now considering how to finance the acquisition. There are two options.

Lease

A vehicle leasing company, National Leasing, has offered the company a lease with annual payments at the beginning of each year over the next 5 years. The lease will require annual payments of R422 800 at the beginning of each year.

Loan

The bank's commercial lending division has offered the company a loan of R1.8m at an annual interest rate of 10%, repayable in arrears in equal annual instalments.

The useful life of the truck is 5 years and the truck is expected to have a residual value equal to one third of its cost. The company's tax rate is 30%. The company's cost of capital is 11%. If the company purchases the truck, the effective life is 5 years and the company will qualify for a depreciation deduction of 20% per year, straight-line. The firm will need to spend R60 000 per year on maintenance, whether the company purchases or leases the truck.

Required:

(a) What is the interest rate implicit in the lease option for Cape Trucking? [*Note*: Use only the pre-tax lease payments.]

(b) What is the annual arrears payment that the company will be required to make on the loan? Prepare an amortisation schedule for the loan option.

(c) Undertake an analysis of the lease and borrow to buy options and determine the net advantage of leasing. Assume that the leasing company is prepared to include maintenance in the above lease contract so that by leasing, the company will not be required to pay for maintenance. Which financing option should Cape Trucking use? Page 15-32

(d) How would the net advantage of leasing (NAL) change if the lease contract permits the company to take ownership of the asset at the end of Year 5 for no consideration? Assume firstly that the company plans to dispose of the asset for its market value. What would happen to the NAL if the company intends to use the asset for a further five years after Year 5, and the residual value is expected to be zero at the end of that period? Assume that the maintenance charge will apply after the lease comes to an end.

Question 15.10

Assume the same facts as 15.9. Let's look at the lease from the perspective of the lessor. Assume that the lessor, owing to purchasing economies, is able to acquire the truck at a discount of 10% of the cost to the company. Further, owing to its position and expertise, National Leasing will be able to achieve a higher residual value than the company and can achieve a premium of 10% above any residual value estimated by the company. The cost of financing for National is 8% per year before tax.

Required:

(a) What is the implicit pre-tax return that National Leasing is able to achieve from the lease?

(b) What is the minimum lease payment that National can set and still break-even from an NPV perspective, after taking tax into account? Use a tax rate of 30% and a depreciation deduction of 20% per year straight-line. Assume that lease receipts are taxable at the same time as the lease receipt at the beginning of the year. What is the minimum lease payment if the lease receipt is taxable a year subsequent to receipt?

Question 15.11

JIT Ltd is a manufacturer of motor vehicle components. The company is required to invest in used tooling equipment at a cost of R50m. A finance company that specialises in the leasing of such equipment has offered either a 5-year lease requiring annual payments of R12m per year payable in advance or a loan which is repayable in a single repayment at the end of 5 years. The leasing option includes servicing and maintenance with a value of R2m per year payable in arrears. The tax rate is 27%. The lease amounts are tax deductible one year after payment as they take place at the beginning of each year. If the company takes the loan, the company will be required to pay for the servicing and maintenance costs. The residual value is expected to be R20m in 5 years' time. The cost of capital is 13% and the before-tax cost of debt is 10.9589%. The company is able to depreciate the cost of the tooling equipment on a straight-line basis over 5 years. To acquire the equipment at the end of the lease, the lessee is required to make a residual payment of R9m. What is the net advantage of leasing?

Question 15.12

Solarwinds Ltd is considering replacing its current computer system and network with a new integrated and wireless system at a cost of R3m and the system is expected to have an effective life of 4 years. The residual value is expected to be 30% of cost. In terms of SARS guidelines the company will be able to depreciate the new system on a straight-line basis over its effective life. Maintenance costs and insurance which are estimated to be R128 000 per year will be paid by the lessor if the company uses lease financing. The lease option will involve annual payments of R780 000, payable at the beginning of each year (four payments).

The company's bank is prepared to offer a loan for the full amount at an interest rate of 9.589% per year, payable in arrears. The company tax rate is 27% and the company's cost of capital is 11%.

Required:

(a) What is the net advantage of leasing? Determine whether the company should lease or borrow and purchase the system.

(b) What is the annual interest deduction for each year if the company purchases the new system? (c) If the company expects to continue to use the equipment for a further 4 years after the initial period, and the company is able to purchase the system from the lessor at the expected residual value, how does this change the net advantage of leasing? Assume that the residual value at the end of the further 4-year period is R300 000. Page 15-33

Question 15.13

Nicrom Ltd is required to replace a conveyor system at a cost of R2.4 million. The company is able to finance the acquisition with either a lease or a loan to purchase the system.

Lease

The company is able to lease the system in terms of a five-year lease which requires annual beginning-of-year payments of R550 000 and the lessor will pay for all maintenance costs which are expected to be R240 000 per year. The company will exercise its option to purchase the system at the end of 5 years at its expected market value of R800 000.

Purchase

The cost can be financed with a five-year bank loan at an interest rate of 8.2192% over 5 years. The loan will be repaid in terms of a single bullet payment at the end of 5 years. The system will qualify for a depreciation deduction of 40% of cost in the first year and 20% of cost in each year for the subsequent three years. The corporate tax rate is 27%. The company will continue to use the system for a further 5 years after the initial 5-year period. The residual value at the end of 10 years is zero.

Required:

(a) Determine the net advantage of leasing by comparing the present value of the cash flows of each financing alternative.

(b) If the loan is repayable in equal annual instalments over 5 years in arrears, what effect will this have on the decision? What is the annual instalment? Show why the present value of the loan will not change.

Question 15.14

Sanele Fishing is able to obtain a 100% loan to finance the acquisition of a new fishing vessel at a cost of R2.4 million. The effective life of the fishing vessel is 10

years and the company will be able to depreciate the fishing vessel on a straight-line basis over 10 years. As an alternative to the borrow and buy option, the company is able to obtain lease finance from a finance company.

The lease will require the company to make an annual lease payment of R480 000 at the beginning of each year for 5 years. Sanele Fishing will make use of the option to acquire the fishing vessel at market value at the time of the last lease payment. The market value of the fishing vessel is expected to be R1.6 million at the end of the lease term, and this will qualify for a depreciation deduction of 20% per year straight-line. The value at the end of its effective life is expected to be R200 000. The estimated maintenance expenses are R120 000 per year. The lessor will pay for any maintenance during the term of the lease. The before-tax cost of debt is 9.589%. The corporate tax rate is 27%.

Required:

(a) Determine whether the company should lease or borrow to buy the fishing vessel.

(b) Comment on whether all cash flows should be discounted at the same rate. In particular, what discount rate should be used to discount the future residual value of the vessel?

Question 15.15

The board of directors of Bluestone Limited has decided to install a computer system to control its manufacturing operations. The financial director's analysis, with the aid of capital budgeting techniques, has established that the savings effected by the new system would result in an internal rate of return which is significantly higher than the company's cost of capital of 13%. The current company tax rate is 27%.

The invoice price of the computer system, including installation, is R3.5m. This amount can be financed by a bank loan at an interest rate of 8.2192% p.a. to be repaid in five equal annual instalments in arrears (interest and capital). If the system is purchased, the manufacturer undertakes to maintain and service the computer for a charge of R220 000 p.a. Instalments are payable in arrears.

The computer manufacturer is also prepared to lease the computer to Bluestone Limited for Page 15-34 R950 000 p.a. on a five-year lease. Included in the lease agreement is a service contract in terms of which the manufacturer undertakes to maintain the equipment in good order. Instalments are payable in arrears.

The computer has an estimated physical life of approximately ten years; at the conclusion of this period the salvage value is expected to be nil. Bluestone Limited has adopted the straight-line method of depreciation as its accounting policy. The SA

Revenue Service has accepted the basis of depreciation and the estimated effective life for tax purposes.

In terms of the company's corporate plan, a new production process with inbuilt control mechanisms is to be introduced in approximately five years which will render the computer redundant. The computer will, however, have a resale value of R1.75m at this stage.

As financial director of Bluestone Limited, the board has requested that you make a recommendation as to whether to finance the system by borrowing the money from the bank at 8.2192% p.a. or by leasing the equipment for five years. There is considerable disagreement within the company as to the discount rate to be employed in evaluating the proposal; the managing director, however, favours the after-tax cost of debt as the discount rate.

Required:

(a) In the capital budgeting decision, it appears that the board of directors of Bluestone Limited paid particular attention to the internal rate of return (IRR) yielded by the project. What are possible reasons for management's preference for the IRR approach? There is a great deal of support for the net present value approach in decisions of this nature; suggest reasons supporting the NPV approach.

(b) Using the discount rate favoured by the managing director, make a recommendation to management regarding the desirable financing method.

(c) Evaluate the arguments for and against the discount rate proposed by the managing director. In your answer consider other possible discount rates.

Question 15.16

The management of Empumalanga Limited are investigating whether to increase plant capacity. Empumalanga is a manufacturing company producing engineering products. One of its products, a Zaphod, has a selling price of R5 500 per unit.

Based on a market survey which cost R5m (and is included in the tax assessed loss), the marketing manager is convinced that sales would increase to the indicated demand of 30 000 units for the next five years if the company was to increase its current capacity.

The company's current capacity is 25 000 units per annum. Each unit has a variable cost of R3 000 and an allocated fixed overhead cost of R1 000, based on capacity. The tax value of the existing plant is R10m and the remaining useful life is expected to be five years. The existing plant was purchased five years ago and has a current market

value of R8m. In five years' time the market value is expected to be zero. The current depreciation charge, which is the only non-cash expense included in fixed overheads, is R2m per annum. The current date is 31 March 20×0, the end of the current financial and tax year. Any recoupment or scrapping allowance on the sale of the existing plant & machinery will affect the company's taxable income one year later in 20×1.

The new plant could be purchased for R72m and has a maximum capacity of 50 000 units. The new plant has an estimated useful life of five years at the end of which the salvage value is expected to be R13m. It is estimated that, because of technological advances, the new plant will result in increased efficiencies and a reduction in variable costs of R750 per unit. Fixed costs, other than depreciation, are expected to remain unchanged.

The new plant qualifies for a depreciation deduction of 40% of cost in the first year and 20% of cost in each of the three subsequent years. The depreciation deduction for the old plant is the same as the accounting depreciation charge. In the past the company had been unprofitable and, as a result, had built up an assessed tax loss of R15m (inclusive of the marketing survey cost of R5m). Profits from the production of other products are expected to result in taxable income of R5m per annum for the next five years before taking into account the profits from producing Zaphods.

The new plant can be financed by way of a loan bearing interest at 8.2192%, being the normal before-tax cost of debt, repayable annually in arrears for five years. Alternatively, a lease is available with instalments of R20m payable annually in advance for five years. The lease payment occurs at the beginning of the year but the deduction from taxable income occurs at the end of each year. Any utilisation of an assessed tax loss from a prior year is limited to 80% of taxable income of any subsequent year.

Assume a company tax rate of 27%. The company's cost of equity is 12% and the debt ratio is 50%.

Required:

Advise management as to whether they should invest in the new plant, assuming their objective is to maximise shareholders' wealth. If the investment is made, what financing alternative should be employed?

Question 15.17

Ukusa Ltd is a large and profitable company which processes and markets a brand of canned fish. The management of Ukusa is keen to expand and is constantly on the

lookout for suitable investment opportunities.

In the past Ukusa have always used an outside firm to do their canning for them at a cost of R15.00 a can. However, a new mini canning plant has just been released on the market and the management of Ukusa are keen to establish whether they would be better off acquiring the use of their own canning plant, rather than using the services of an outside firm.

The canning plant will cost R24 million and will qualify for a depreciation deduction of 40% of cost in the first year and 20% of cost in each of the three subsequent years. The plant's useful life is expected to be five years. At the end of five years the plant is expected to have a residual value of R4 million, although this residual value will not be taken into account in computing depreciation. It is estimated that 720 000 cans will be processed every year for the five years. A compulsory maintenance contract will cost R800 000 per annum, while other expenditure, including depreciation, will amount to R11.50 a can.

In addition to the special introductory price of R24 million, the manufacturer is offering two special introductory financing packages:

- a loan repayable in five equal annual arrear instalments is available at 5.4795%;
- a lease with five annual rentals of R5.2 million, payable annually in advance, is available.

The lease rentals include the maintenance contract. If the plant is leased, ownership of the asset will not pass to the lessee at the end of the lease.

You may assume a company tax rate of 27%. Ukusa has a before-tax cost of debt of 10.9589% and a weighted-average cost of capital of 14%. The company is able to utilise any tax deductions from the project as UKUSA has significant taxable income from product sales. Deductions are not ringfenced to individual projects.

Required:

(a) Employing a net present value analysis, advise management whether they should proceed with the investment in the canning plant.

(b) Calculate the net present cost of both the lease and the 5.4795% debt alternative.

(c) Calculate the net advantage of the special financing packages and comment on the impact that your result has on the advice given in (a) above.

Question 15.18

Sites Ltd, a property owning company, has recently purchased an industrial property for R9m on which it has erected a multi-purpose building which cost R52m. The company is considering two lease proposals and requests your advice.

(i) Warehouses (Pty) Ltd is prepared to pay a lease premium of R7.8m and an annual rental of 11% on the capital cost of the property (excluding land) and will use the property to store goods for its holding company which is engaged in shipping.

(ii) Factories (Pty) Ltd requires certain alterations to be made to the building in order to make it Page 15-36 suitable for manufacturing operations. These will cost R8m but Factories (Pty) Ltd will make available a loan of R8m to Sites Ltd at a before-tax interest rate of 5.4795% per annum to finance the alterations. The annual rental will be determined at 12% on the capital cost of the property including alterations. The loan will be repaid at the end of the lease.

Both prospective tenants require ten-year leases which they will not necessarily renew. The alterations will not enhance the value of the property. The risk of either company defaulting is extremely low and both companies have very high credit risk ratings. The company will evaluate both leasing options by using a discount rate equal to the after-tax interest rate of Sites. The residual values are not relevant as they will apply to both options and the company is not intending to dispose of the property. The market value of the building is expected to equal its tax value at the end of 10 years and there is no residual payment for either option. Sites Ltd employs loan finance at a rate after tax of 8% per annum and uses any favourable cash flow to reduce borrowings. Sites Ltd has been making substantial profits for a number of years. The rate of normal tax is 27%. The company will qualify for an annual allowance of 5% of the cost of construction and the cost of any alterations if the building is used for manufacturing purposes.

Required:

Determine which lease is the most favourable to Sites Ltd. You may assume that all receipts and payments are made on the last day of each year except for the lease premium or payment for the building alterations which will be made on the first day of the first year.

(*SAICA ITC/QE – adapted*)

Question 15.19

The board of directors of Union Ikapa Manufacturing (Pty) Ltd was considering whether or not to invest in a pre-owned cutting machine which would increase existing capacity.

As the marketing manager had often stressed the need for increased capacity in the cutting plant, the financial manager decided to investigate this proposal. There did not appear to be any other suitable investment opportunity. The financial manager performed a net present value analysis on the anticipated cash flows of the project by discounting it at the company's weighted- average cost of capital. This resulted in a negative net present value amounting to R1.647m and so a recommendation was forwarded to the general manager not to invest.

On hearing that the board of directors of Union Ikapa had decided not to go ahead with the project, the supplier of the cutting plant, in order to make the deal more attractive, suggested that, while no further discounts on the purchase price could be offered, competitive finance could be arranged which would in effect reduce the cost of the project. Two alternatives were proposed:

- A R50m loan to cover the purchase price of the machine was offered at 9.589%, with interest payable annually in arrears and capital repayable at the end of five years.
- The plant could be leased for five years at R10m per annum, payable annually in advance.

If the cutting machine is purchased outright, service and maintenance charges will be R500 000 per annum. The lease charge includes maintenance and servicing. The salvage value of the machine five years hence is expected to be R3m. The company is allowed a depreciation deduction of 20% per annum, straight-line. The cutting machine has an estimated useful life of five years.

The financial manager was adamant that, regardless of the above proposals, the company should still not invest as a negative net present value would result in a decrease in shareholders' wealth. He also believed it necessary to keep the investment and financing decisions separate as they were two separate issues.

The company has an after-tax cost of debt of 8% and a weighted-average cost of capital of 12%. Irrespective of the financing method used, the weighted cost of capital is not expected to change significantly. The rate of company tax is 27%. All cash flows relating to expenses (other than the lease payments), interest and taxation should be assumed to occur at the end of the respective year. The company will have a large taxable income for the foreseeable future.

Required:

(a) Discuss the financial manager's views that financing implications should be ignored in this case; and

(b) Calculate which financing alternative is most favourable, and conclude whether or not the investment should be made.

(*SAICA ITC/QE – adapted*)

Question 15.20

(i) It has been suggested that a lessee should evaluate a leasing decision on the net present value basis using the company's marginal borrowing rate. A lessor, on the other hand, should evaluate leasing projects using the company's weighted average cost of capital.

Required:

(a) Comment on the validity of these two suggestions.

(b) Outline and explain the alternative approaches to lease evaluation available to a lessor.

(ii) A lessor company is considering a transaction under which it would acquire equipment at a cost of R102m including expenses and would lease it to a client for five years, at the end of which time the equipment would have no residual value. The cost of funds to the lessor company is 8% per annum before tax, interest being payable half-yearly, the first payment being six months from the date of acquisition of the equipment. The reinvestment rate on surplus funds is forecast at 5%, receivable half-yearly less tax. The company would be entitled to 100% first- year depreciation deduction on the total cost of the equipment. The rate of corporation tax is expected to be 30% throughout the life of the contract. The company's accounting period ends on 30 June and corporation tax is payable on 1 January, 18 months after the end of the accounting period to which it relates. The acquisition under consideration takes place on 1 January, 20.4. The suggested rental terms are R12m each half-year, payable in advance.

Required:

Make detailed calculations to show whether or not the proposal would be financially justifiable.

(*CIMA*)

Question 15.21

The finance director of Tsoela Ltd wishes to know whether to acquire a pre-owned machine. The machine would either be purchased for cash, or on hire-purchase (HP), or would be leased through a finance lease agreement. The purchase cost of the machine is R14.25m. It has an expected life of four years and, if purchased, will be sold at the end of year four. The realisable value at the end of four years is expected to

be R2.25m. The machine is expected to generate before-tax net cash flows of R5.63m per year for four years. If hire-purchase is used, an initial 20% deposit is payable, followed by four equal annual year-end payments covering both the interest and principal elements of the payment. The interest rate on the hire purchase loan is 10%. Leasing would involve an annual rental payment of R3.9m, payable in advance.

The company pays tax at the rate of 27% per year, in arrears, and a depreciation deduction of 20% per year is available on the machinery.

Tsoela Ltd would finance the outright purchase of the machine with a loan repayable in full after four years. The 20% initial deposit on the hire-purchase deal would be financed in the same way. The company's normal before-tax cost of debt is estimated to be 10.9589% per year and its cost of equity is 14% per year. The risk associated with the current project is similar to that of the company's other activities. A debt ratio of 50% is maintained. Page 15-38

Required:

(a) Discuss the relative advantages to the lessee or borrower of leasing and hire-purchase.

(b) Evaluate whether the machine should be installed and, if so, which means of financing should be chosen.

State clearly any assumptions that you make.

Question 15.22

Cape Instruments Ltd is a listed company on the JSE. The company was founded in the early 1990s when the management identified a market for medical devices and fitness and exercise equipment.

The management of Cape Instruments Ltd are investigating whether to establish a local manufacturing plant in order to manufacture a new fitness machine called a 'Sunfit 1800'.

Based on a market survey completed on 1 January 20.2 at a cost of R1m, the probable 'Sunfit 1800' unit demand for the next four years has been estimated to be as follows:

	Expected demand per annum	Probability
High	12 000	30%
Medium	10 000	40%
Low	8 000	30%

The demand between years is considered to be dependent, i.e. if there is low demand in Year 1 it is expected that subsequent years will reflect low demand, while if the

demand was high in Year 1, then subsequent years will reflect high demand.

New plant with a maximum capacity of 10 000 units can be purchased for R30m. The plant has an estimated useful life of four years at the end of which the salvage value is expected to be R2.5m. It is estimated that each unit of 'Sunfit 1800' will incur variable costs of R10 000, while fixed overhead costs based on a capacity of 10 000 units and before allowing for depreciation, are expected to be R2 500 per unit. These fixed overhead costs will be avoidable if local manufacture does not take place. The 'Sunfit 1800' will sell at R15 950, some R5 000 less than the imported machines. This will mean that the current sales of the imported machines will drop from 7 000 per annum to zero. The contribution generated on each of these imported machines amounts to R2 500 per unit, while fixed costs amount to R17.5m per annum. However, these fixed costs will still be incurred if local manufacture takes place.

The new plant would be subject to a 20% straight line depreciation deduction. The company has built up an assessed loss of R3m as at the financial year ended 31 December 20.1. Profits from the production of other products are expected to result in taxable income of R1m per annum for the next four years.

The new plant can be financed by way of a loan bearing interest at 10.9589%, being the normal before-tax cost of debt. The loan would be repaid in one instalment at the end of the fourth year, while interest is payable annually in arrears on the amount outstanding. Alternatively, a lease is available with instalments of R10m payable annually in advance for four years. At the end of the lease the asset will become the property of the lessee. Once the company achieves taxable income, a company tax rate of 27% will apply. The company has consistently retained a debt ratio of 50% and its cost of equity is estimated to be 14%. Any utilisation of the tax loss is limited to 80% of the taxable income (excluding the loss) in any year but can be carried forward to future years. Page 15-39

Required:

(a) Advise management whether they should consider local production and invest in the new plant if they wish to maximise shareholders' wealth.

(b) Advise management as to the most appropriate form of finance assuming an investment in the new plant.

(c) Discuss the impact that inflation would have on your analysis above.

(*SAICA ITC/QE – adapted*)

Question 15.23

Perfect Home Panels (PHP) is a division of a major industrial engineering company. The division has been set up to manufacture and deliver panels that will form the external structures and walls to be employed in low-cost housing. The government has set up bidding processes for firms to become involved in the provision of these panels and the winner of these contracts will be selected on the basis of competence in terms of product delivery schedules, quality and price. Further, the government will evaluate the social responsibility policies of the bidding firms before granting contracts. PHP is confident that it has amply met the social responsibility requirements of the government. However, the company needs to decide how it will price the full panel units. Each unit will consist of all the external panels required to construct a housing unit including the roof.

The provision of low-cost housing requires the use of materials that provide maximum strength and the use of reinforced materials and multilayered structures. The units should be modularised which allows rapid construction of housing units and modularisation would also promote standardisation, allowing unit expansion as well as facilitating transportation of the units to sites. The materials would be insulated to allow minimum heat loss during the winters while at the same time providing the ability to exclude heat during the summer months.

The durability of the materials is important, particularly regarding consistency and UV stability. The heat efficiency of the units means that it will result in lower heating bills. The company's experience in the use of reinforced materials, particularly in relation to the provision of housing at mining sites, makes the company ideally placed to provide housing in rural areas and in informal urban areas. The government will subcontract builders to install the units at sites provided by government.

The government requires that 1 000 units be produced per annum and delivered to specified sites in five rural regions in the Northern Cape. The contract will run for 5 years and will not be renewable. There are two alternative processes that can be employed to meet the design specifications for the housing units. In terms of engineering specifications, these processes are equal in terms of providing a standard product. Therefore the company needs to choose the process that will allow the firm to submit the lowest per unit bid. The company's marginal tax rate is 27% and the company wishes to achieve its required return of 15% per annum after tax on this project.

The first alternative (P1) requires high-pressure moulding that increases the strength of the unit. The plant and machinery to manufacture the units will cost R23 million

and this qualifies for the depreciation allowance of 20% per annum straight-line. The company has sufficient income from other product lines to fully utilise all allowances. The residual value of the plant and machinery is expected to be zero at the end of five years.

The second alternative (P2) will use corrugated inner sections. The process requires investment in equipment costing R12 million. The economic life of this equipment for this purpose is 5 years. However, the residual value of the equipment at the end of 3 years is expected to be R6 million. The machinery can be replaced at the end of 3 years at an expected cost of R12 million so that PHP can continue production. The residual value of the replacement machine at the end of a further 2 years is expected to be R8 million. The plant and equipment qualifies for the depreciation allowance of 20% per annum, straight-line.

If PHP holds on to the initial plant and equipment for the full 5 years, then the residual value is expected to be only R3 million at the end of the 5-year period.

The second process requires the purchase of the inner sections and this adds to the cost of materials required to undertake this process. Page 15-40

The total before-tax operating costs (R millions) per process for 1 000 units will amount to the following:

	P1	P2
Materials	7.6	13.2
Labour	1.0	1.8
Fixed cost	9.45	7.0
Rent	0.8	0.8

The fixed costs exclude depreciation and represent incremental cash costs. If PHP selects the second process and decides to hold on to the plant and equipment for the full 5-year period, then the fixed costs will increase to R7.34 million per annum in Years 4 and 5 as a result of the maintenance required to maintain the complex equipment in the condition required to produce the requisite number of units.

PHP is to finance the equipment by the use of a loan or the use of a lease. The equipment for Process 1 can be leased from the financial institution that offered the loan finance. The lease payments would amount to R5.85 million payable annually in advance. The manufacturer of the equipment to be used in Process 2 is prepared to offer PHP asset-specific lease finance. However, in terms of this lease option, PHP can choose to either pay R2.79 million annually in advance or R2.89 million annually in arrears. The manufacturer is able to offer this lease option even if PHP decides to

replace the equipment after three years. The lease instalments will remain the same over the 5-year period and cover the maintenance incurred when equipment is not replaced in Years 4 and 5.

PHP is able to raise loan finance at its normal after-tax cost of debt of 8% per annum. Loans can be repaid and drawn as required in terms of the various purchase options.

The loan finance is repayable when the plant is sold. However, PHP can obtain additional loan finance at the same rate, if the plant is replaced during its economic life.

Required:

(a) Based on net present costs, which production process should PHP utilise to produce the units?

Note: ignore any effect for now arising from whether the firm leases or borrows and purchases the equipment.

(b) At what price per unit should PHP submit a bid to the government authorities?

(c) Determine on the basis of net present costs whether the company should lease or borrow and purchase the machinery used in the production process selected in (a). Indicate whether the bid price will be affected by the financing method selected and calculate the net effect.

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[1] The actual numbers are 4 217 066.6 – 4 100 197.4 = 116 869.2

[2] You can use the Excel function, =PMT(0.09,5,50000000,0,) or apply a financial calculator to determine the PMT of R12 854 623.

N	I/YR	PV	PMT	FV
5	9.0%	50,000,000	-12,854,623	0